Financial Analyst & FP&A — Real Job Simulation

Real Job Simulation — Do the Actual Job, End to End (India, FY 2026-27)

One continuous company case: budgeting, variance, MIS, forecasting, 3-statement model, DCF, Power BI

Contents

Your First Week in FP&A: The Business, the Model, and the Data

The ask

It's Monday, 6 April 2026. You've just joined Nirvana Traders & Services Pvt Ltd ("NTSPL"), Hyderabad, as the FP&A Analyst reporting to the CFO, Mrs. Rao. Ten minutes into your first 1:1 she says:

> "Before you touch a single formula, I need you to *understand the business*. By Friday I want a one-page map — how we make money, where the numbers live, and a walkthrough of the budget model you're inheriting. If you can't explain our revenue on the back of an envelope, you can't forecast it."

Deadline: Friday 10 April 2026, 5 pm. No new model. Just: learn the machine before you drive it.

What you're given

A shared drive folder (`\\ntspl-fs01\Finance\FY27`), read access to Tally Prime, and a login to the sales CRM. The CFO drops three artifacts on your desk.

1. The FY2026-27 approved budget (one-line summary the board signed off):

Line	FY2026-27 Budget
Revenue	Rs 12.00 cr
— Goods	Rs 9.00 cr
— Services (AMC/install)	Rs 3.00 cr
COGS	Rs 8.40 cr
Gross profit (30%)	Rs 3.60 cr
Employee cost	Rs 1.08 cr
Other opex	Rs 0.78 cr
Depreciation	Rs 0.144 cr
EBIT	Rs 1.596 cr
Finance cost	Rs 0.09 cr
PBT	Rs 1.506 cr
PAT (~25%+cess)	~Rs 1.11 cr

2. The revenue drivers behind those two lines:

Segment	Driver	Volume	Rate	Revenue
Goods	units x ASP	90,000 units	Rs 1,000	Rs 9.00 cr
Services	AMC contracts x price	250 contracts	Rs 1,20,000	Rs 3.00 cr

3. Where the data lives (the CFO's scribbled system map):

System	Owner	Holds	Refresh
Tally Prime (ERP)	Accounts (Suresh)	GL, actual P&L, AP/AR, inventory	Daily; locked at month close
Sales CRM	Sales head (Kiran)	Orders, units, ASP, AMC pipeline	Live
HR sheet (Excel)	Admin (Latha)	Headcount, CTC, joining dates	Monthly
Bank portal	CFO	Cash, term-loan balance	Daily

Build it — step by step

Your Friday deliverable is understanding, not a model — but the way you *acquire* that understanding is by reverse-engineering the inherited Excel file, `NTSPL_Budget_FY27.xlsx`.

Step 1 — Map the money. Two revenue engines. Goods is a **volume x price** business: 90,000 units x Rs 1,000. Services is a **contract count x annual value** business: 250 AMC x Rs 1,20,000. Memorise the identity:

```
Revenue = (Goods units x ASP) + (AMC contracts x contract value)
        = (90,000 x 1,000) + (250 x 1,20,000)
        = 9,00,00,000 + 3,00,00,000 = 12,00,00,000  (Rs 12.00 cr)
```

Step 2 — Learn the margin shape. Goods carry 25% gross margin, services 45%. Blended it lands at 30% only *because of the mix* (75% goods / 25% services). Prove it:

```
GP = 9.00cr x 25% + 3.00cr x 45% = 2.25cr + 1.35cr = 3.60cr  → 30% of 12.00cr
```

Write that down — the day the mix shifts, the "30%" moves even if nothing else changes. This is the single most misunderstood number in the company.

Step 3 — Open the model and label every tab. A clean FP&A workbook follows the **inputs → calcs → outputs** discipline. You colour-code the tabs:

Tab (colour)	Type	What's on it
`Assumptions` (green)	Input	Drivers: units, ASP, contracts, margins %, headcount plan, tax rate. The *only* place you type numbers.
`Rev_Build` (grey)	Calc	Revenue by segment, by month, from Assumptions
`Opex_Build` (grey)	Calc	Employee cost from HR roster; other opex
`P&L` (blue)	Output	Monthly + annual P&L, budget column
`Actuals` (grey)	Calc	Tally GL dump, mapped to P&L lines
`Variance` (blue)	Output	Budget vs actual bridge
`Dashboard` (blue)	Output	KPI tiles + charts

The golden rule you confirm holds: **no hard-coded numbers inside calc or output tabs — every figure traces back to a green Assumptions cell.** You test one: click the goods-revenue cell on `Rev_Build` and it reads `=Assumptions!B4*Assumptions!B5`, not `=9000000`. Good — the model is a machine, not a picture of a machine.

Step 4 — Trace one number end to end. You pick employee cost. `Opex_Build` sums the HR roster with `=SUMPRODUCT(CTC_range, active_flag)` and phases it monthly; the annual total feeds `P&L!EmployeeCost` which reconciles to the Rs 1.08 cr budget line. You now know the *path*, so when it breaks you'll know where to look.

Step 5 — Fix the KPI set. From the CFO's board deck you extract the six numbers she actually reports:

KPI	Definition	FY27 budget anchor
Revenue	Goods + services	Rs 12.00 cr
Gross margin %	GP / revenue	30%
EBITDA	EBIT + depreciation	Rs 1.74 cr
DSO	Debtors / revenue x 365	60 days
Headcount	Active employees	15 → 18 by Q4
Cash balance	Bank + cash	Rs 35 lakh (opening)

The deliverable

You produce a **one-page Business & Data Map** and a two-minute model walkthrough. The one-pager:

NTSPL on one page (FY2026-27) - **Who we are:** Hyderabad trader of industrial electrical components (goods) plus installation & AMC (services). - **How we make money:** Goods = 90,000 units x Rs 1,000 = Rs 9.0 cr (25% GM). Services = 250 AMC x Rs 1.2 lakh = Rs 3.0 cr (45% GM). Total Rs 12.0 cr, blended GM 30%. - **What flows to the bottom:** GP 3.60 cr → less opex 1.86 cr, depreciation 0.144 cr → EBIT 1.596 cr → less finance cost 0.09 cr → PBT 1.506 cr → PAT ~1.11 cr. - **Where the data lives:** Actuals in Tally (locked at close); drivers in CRM; people in the HR sheet; cash on the bank portal. - **The model:** one workbook, Assumptions → builds → P&L/Variance/Dashboard. Green = type here, everything else is formulas.

Commentary (analyst voice): "NTSPL is a two-engine business. Goods drives scale and cash-cycle risk; services drives margin. The blended 30% is a mix artefact, not a floor — my forecasts have to model the two engines separately or the margin will lie to us."

How it's reviewed

The CFO checks three things. **One — can you explain revenue without opening Excel?** She'll ask "what's our goods revenue and why" and expect "90,000 units at a thousand rupees, nine crore." **Two — do you understand the margin mechanics**, i.e. that 30% is mix-driven, not a constant. **Three — do you know where each number is born**, so month-end doesn't become an archaeology dig. She'll spot-check by asking "where does employee cost come from?" and expect "HR roster, SUMPRODUCT of CTC times active flag, on Opex_Build."

Common mistakes & red flags

- **Treating the model as a picture.** Hard-coding a number into an output tab breaks the machine silently. Every output cell must trace to green.
- **Averaging the margin.** Saying "our margin is 30%" and applying it to any revenue mix. Goods and services must be modelled separately.
- **Not knowing the data lineage.** If you can't name the source system for a P&L line, you can't defend the number when it moves.
- **Confusing EBIT and EBITDA.** EBIT is 1.596 cr; add back depreciation 0.144 cr for EBITDA 1.74 cr. Interviewers love this trip-wire.
- **Ignoring the cash cycle on day one.** A goods-heavy trader lives and dies by DSO/DIO/DPO; note it early even if you're not modelling it yet.

On the job & in the interview

The "why": FP&A exists to connect *drivers* (units, contracts, people) to *outcomes* (P&L, cash) so the business can steer, not just report. Your first-week job is to internalise the drivers so every later forecast is a lever pull, not a guess.

Jargon to own: **driver-based**, **inputs/calcs/outputs**, **single source of truth**, **data lineage**, **mix effect**, **tie-out**.

Q: "Walk me through NTSPL's business model in 60 seconds." A: "Two engines. Goods — 90,000 units at Rs 1,000 ASP, Rs 9 cr, 25% gross margin — is a volume-and-price trading business with a working-capital tail. Services — 250 AMC contracts at Rs 1.2 lakh, Rs 3 cr, 45% margin — is recurring and high-margin. Blended revenue Rs 12 cr, gross margin 30%, EBIT Rs 1.6 cr, PBT Rs 1.5 cr. The strategic lever is shifting mix toward services to lift blended margin."

Q: "Why is the blended gross margin 30% and not the average of 25% and 45% (35%)?" A: "Because it's revenue-weighted, and goods is three-quarters of revenue. 0.75x25% + 0.25x45% = 30%. If services grew to half the mix, blended margin would rise to 35% with no change in either segment's margin — that's why I forecast the segments separately."

Q: "What would you check in an inherited model first?" A: "Structure and lineage. I confirm inputs/calcs/outputs separation, that outputs contain no hard-codes, and I trace one number — say employee cost — from the HR roster through the build to the P&L line and reconcile it to the Rs 1.08 cr budget. If that path is clean, I trust the machine."

Building the Annual Budget from Drivers

The ask

It's Tuesday, 3 February 2026. Budget season. The CFO forwards you the board's top-down target — "Rs 12 crore, hold PBT above Rs 1.5 crore" — and says:

> "Build it bottom-up from drivers. I don't want a number I can't defend line by line to the board. Units, price, contracts, headcount — every rupee has to hang off an assumption I can argue with. Draft P&L on my desk Thursday, monthly phasing included."

Deadline: Thursday 5 February 2026. The output must reconcile *exactly* to Rs 12.00 cr revenue and Rs 1.506 cr PBT — that's the deal the board signed.

What you're given

The CFO's driver assumptions for FY2026-27, plus last year's actuals for sanity:

Driver assumptions (FY2026-27):

Driver	Value
Goods units (annual)	90,000
Goods ASP	Rs 1,000
Goods gross margin	25%
AMC contracts	250
AMC contract value	Rs 1,20,000
Services gross margin	45%
Opening headcount	15
Closing headcount (Q4)	18
Avg fully-loaded cost/head	~Rs 6.4 lakh/yr
Other opex	Rs 0.78 cr
Depreciation	Rs 0.144 cr
Term loan	Rs 1.20 cr @ 9.5%
Tax	25% + 4% cess

Seasonality (from CRM history): goods are back-loaded — H1 ~45%, H2 ~55%, with a March spike. Services (AMC) are booked evenly, 1/12 per month.

Build it — step by step

Step 1 — The driver block (green Assumptions tab). Lay drivers in labelled cells, never inside formulas:

Cell	Label	Value
B4	Goods units	90,000
B5	Goods ASP	1,000
B6	Goods GM %	25%
B8	AMC contracts	250
B9	AMC value	1,20,000
B10	Services GM %	45%

Step 2 — Revenue build. On `Rev_Build`:

```
Goods revenue    = B4*B5      = 90,000*1,000     = 9,00,00,000
Services revenue = B8*B9      = 250*1,20,000      = 3,00,00,000
Total revenue                                     = 12,00,00,000
```

Step 3 — Monthly phasing. Store a 12-column seasonality vector (`Phasing!C2:N2`) that sums to 100%, then spread each segment. Goods uses the back-loaded curve; services is flat. Monthly goods revenue:

```
=$B$4*$B$5 * Phasing!C$2     ' dragged across Jan..Dec, phasing row sums to 1
```

To validate the spread ties back, cross-foot with SUMPRODUCT — the sum of monthly must equal the annual driver product:

```
=SUMPRODUCT(Phasing!C2:N2, MonthlyUnits!C2:N2)  ' must return 90,000
```

If you had per-SKU or per-region volumes in a table, you'd aggregate with `=SUMIFS(Units[Qty], Units[SKU], "*", Units[Month], C$1)` rather than a single cell — the SUMIFS/SUMPRODUCT pattern is the workhorse for driver roll-ups.

Step 4 — COGS and gross profit. COGS is revenue minus segment GP:

```
Goods COGS    = 9.00cr * (1-25%) = 6.75cr
Services COGS = 3.00cr * (1-45%) = 1.65cr
Total COGS    = 8.40cr
Gross profit  = 12.00 - 8.40    = 3.60cr   (30% blended)
```

Step 5 — Opex build. Employee cost is driven off the **headcount roster with joining dates**, not a flat number. 15 heads at start, +3 phased into Q4. Roughly:

```
Employee cost = SUMPRODUCT(cost_per_head_monthly, active_months)
              ≈ 1.08 cr   (15 heads all year + 3 heads for ~Q4 only)
```

The phasing matters: three hires landing only in Q4 add far less than three full-year heads — the roster method captures that automatically. Other opex Rs 0.78 cr is spread by activity (rent flat, freight follows goods volume).

Step 6 — Down to PBT. Stack the P&L:

```
Gross profit                     3.60 cr
less Employee cost              (1.08)cr
less Other opex                 (0.78)cr
less Depreciation               (0.144)cr
= EBIT                           1.596 cr
less Finance cost (1.2cr*9.5%≈)  (0.09)cr
= PBT                           1.506 cr
less Tax (~25%+cess)           (~0.396)cr
= PAT                          ~1.11 cr
```

Finance cost checks out: Rs 1.20 cr term loan x 9.5% = Rs 11.4 lakh gross, but the average outstanding balance across the year (loan amortising) lands the P&L charge at ~Rs 9 lakh — matching the anchor.

The deliverable

NTSPL Budget P&L — FY2026-27

Line	Amount (Rs cr)	% of revenue
Goods revenue	9.000	75.0%
Services revenue	3.000	25.0%
Total revenue	**12.000**	**100.0%**
Goods COGS	(6.750)	
Services COGS	(1.650)	
Gross profit	**3.600**	**30.0%**
Employee cost	(1.080)	9.0%
Other opex	(0.780)	6.5%
Depreciation	(0.144)	1.2%
EBIT	**1.596**	**13.3%**
Finance cost	(0.090)	0.8%
PBT	**1.506**	**12.6%**
Tax (~26%)	(0.396)	
PAT	**~1.110**	**9.3%**

Quarterly phasing (revenue, Rs cr):

	Q1	Q2	Q3	Q4	FY
Goods	2.10	1.95	2.10	2.85	9.00
Services	0.75	0.75	0.75	0.75	3.00
Total	**2.85**	**2.70**	**2.85**	**3.60**	**12.00**

Commentary: "Revenue is defensible driver-by-driver: 90,000 units and 250 AMCs. Q1 is budgeted at Rs 2.85 cr; the March-heavy goods curve loads Q4 to Rs 3.6 cr, so we can't judge the year on H1 run-rate. PBT lands at Rs 1.506 cr — 6 bps of headroom over the board's Rs 1.5 cr floor, which is thin. The three Q4 hires are the biggest discretionary lever if we need to protect PBT."

How it's reviewed

The controller runs **tie-outs**: does revenue foot to Rs 12.00 cr, does the segment split match the drivers, does GP hit exactly 30%, does PBT clear the Rs 1.5 cr floor? She checks the **phasing sums to the annual** (monthly SUMPRODUCT = annual driver). She stress-tests one driver: "drop ASP to Rs 950 — show me PBT" and expects the model to flow it through instantly because ASP is a single green cell. Finally she checks finance cost against the loan schedule and tax at the right rate.

Common mistakes & red flags

- **Plugging revenue to hit the target.** Top-down "make it Rs 12 cr" with no unit/price logic. The board will ask "how many units?" and you'll have nothing.
- **Flat 30% margin on total revenue.** Skips the segment mechanics; breaks the moment mix moves.
- **Full-year cost for part-year hires.** Costing 18 heads all year overstates employee cost by lakhs. Phase by joining date.
- **Phasing that doesn't reconcile.** A seasonality curve that doesn't sum to 100% quietly changes the annual total. Always cross-foot monthly back to the driver.
- **Forgetting cess.** Tax is 25% *plus 4% cess* ≈ 26%, not a clean 25%.

On the job & in the interview

The "why": a driver-based budget is *defensible* and *flexible* — every number answers "why" (units, price, heads) and every scenario is a one-cell change. A hard-coded budget is neither.

Jargon: **driver-based / bottom-up**, **phasing / seasonality curve**, **cross-foot**, **flow-through**, **PBT floor**, **fully-loaded cost per head**.

Q: "How do you build a revenue budget for a business like this?" A: "Bottom-up from drivers. Goods is units x ASP — 90,000 x Rs 1,000 = Rs 9 cr. Services is contracts x value — 250 x Rs 1.2 lakh = Rs 3 cr. Then phase each by its own seasonality, cross-foot the monthly back to the annual driver, and layer segment margins to get GP. Total Rs 12 cr, GP Rs 3.6 cr."

Q: "Why model headcount cost off a roster instead of a single number?" A: "Because timing changes the cost. Three hires landing in Q4 add roughly a quarter of their annual cost this year, not the full amount. A roster with joining dates and SUMPRODUCT captures that; a flat number overstates employee cost and understates PBT."

Q: "Your PBT is Rs 1.506 cr against a Rs 1.5 cr floor. Comfortable?" A: "Not very — that's under half a percent of headroom. I'd flag the sensitivity: a Rs 50 drop in ASP costs Rs 45 lakh of revenue and, at 25% margin, Rs 11 lakh of GP, which alone breaches the floor. My mitigation lever is deferring or staggering the Q4 hires, worth roughly Rs 5-10 lakh."

Actuals Arrive: Variance Analysis and Commentary

The ask

It's Wednesday, 8 July 2026. Q1 (Apr-Jun) is closed in Tally. The CFO pings you at 9:10 am:

> "Q1 revenue came in at Rs 2.70 crore against Rs 2.85 crore budget — we're Rs 15 lakh light, minus 5%. The board call is Friday. I need a variance bridge that tells me *why*: is it price, volume, mix, or margin? And two paragraphs of commentary I can read out. Don't just tell me we missed — tell me what it means for the full year."

Deadline: Thursday 9 July, noon. This is the deliverable that makes or breaks your reputation — anyone can subtract two numbers; FP&A explains the gap.

What you're given

Q1 budget vs actual dump (from Tally + CRM):

Metric	Budget Q1	Actual Q1
Goods units	22,500	21,000
Goods ASP	Rs 1,000	Rs 1,020
Goods revenue	Rs 2.25 cr	Rs 2.142 cr
Services revenue	Rs 0.75 cr	Rs 0.75 cr
Total revenue	**Rs 2.85 cr**	**Rs 2.70 cr**
Gross margin %	30.0%	28.5%
Opex	(budget)	+Rs 5 lakh over

Note: goods actual revenue = 21,000 x Rs 1,020 = Rs 2,14,20,000 ≈ Rs 2.142 cr. With services flat at Rs 0.75 cr, total = Rs 2.892 cr — but the reported actual is Rs 2.70 cr. The residual Rs 19 lakh is a small returns/credit-note and cut-off adjustment booked to goods; for the price-volume bridge we decompose the goods gap using the clean unit and ASP figures and reconcile the remainder as an "other/adjustment" bucket.

Build it — step by step

The core tool is the **price-volume-mix (PVM) decomposition**. For the goods segment, the total goods variance splits cleanly into a volume effect and a price effect.

Step 1 — Set the identity. Revenue = Units x Price. The variance between actual and budget decomposes as:

```
Volume variance = (Actual units − Budget units) × Budget price
Price variance  = (Actual price − Budget price) × Actual units
```

(Using budget price for volume and actual units for price is the standard convention so the two pieces sum exactly to the total variance — no cross-term left over.)

Step 2 — Volume effect (goods).

```
Volume var = (21,000 − 22,500) × Rs 1,000
           = (−1,500) × 1,000
           = −Rs 15,00,000   (unfavourable Rs 15 lakh)
```

We sold 1,500 fewer units; at the budgeted Rs 1,000 each, that's Rs 15 lakh of lost revenue.

Step 3 — Price effect (goods).

```
Price var = (Rs 1,020 − Rs 1,000) × 21,000
          = 20 × 21,000
          = +Rs 4,20,000   (favourable Rs 4.2 lakh)
```

We charged Rs 20 more per unit on the 21,000 units actually sold — a Rs 4.2 lakh cushion.

Step 4 — Net goods variance.

```
Net goods = −15,00,000 + 4,20,000 = −Rs 10,80,000
```

Check: budget goods 2.25 cr − actual clean 2.142 cr = −Rs 10.8 lakh. Ties exactly.

Step 5 — Services and mix. Services landed on plan (Rs 0.75 cr), so no volume/price variance there. **Mix effect**: because goods (25% GM) fell and services (45% GM) held, the *share* of high-margin services rose — that partially props up blended margin, but the absolute goods shortfall still drags total revenue.

Step 6 — Margin and opex bridges. Gross margin came in at 28.5% vs 30.0% budget — 150 bps light. On Rs 2.70 cr actual revenue that's a gross-profit drag of:

```
Margin var = 2.70cr × (28.5% − 30.0%) = 2.70cr × (−1.5%) = −Rs 4.05 lakh
```

Driven by input-cost inflation on goods plus the mix/return noise. Opex ran Rs 5 lakh over budget (early hires pulled forward from the Q4 plan).

Step 7 — Reconcile the headline. Revenue bridge from budget Rs 2.85 cr to actual Rs 2.70 cr:

```
Budget revenue                     2.850 cr
Goods volume        −0.150 cr
Goods price         +0.042 cr
Other/returns adj   −0.042 cr
Services            0.000 cr
= Actual revenue                   2.700 cr
```

The deliverable

Q1 FY27 Revenue Variance Bridge (waterfall)

Step	Rs cr	Running total
Budget revenue		2.850
Goods volume (−1,500 u × Rs 1,000)	(0.150)	2.700
Goods price (+Rs 20 × 21,000 u)	0.042	2.742
Other / returns & cut-off adj	(0.042)	2.700
Services (on plan)	0.000	2.700
Actual revenue		**2.700**

Gross-profit bridge:

Step	Rs lakh
Budget GP (Q1, 30% × 2.85cr)	85.5
Revenue shortfall effect (30% × −15L)	(4.5)
Margin-rate effect (−1.5% × 2.70cr)	(4.05)
Actual GP (≈28.5% × 2.70cr)	**≈76.95**

Commentary (CFO-ready):

> "Q1 revenue of Rs 2.70 cr is Rs 15 lakh (−5%) below the Rs 2.85 cr budget. The miss is a **volume problem, not a pricing problem**: we shipped 21,000 units against 22,500 planned — a Rs 15 lakh volume drag — only partly offset by Rs 4.2 lakh of favourable pricing, as ASP held at Rs 1,020 versus Rs 1,000 budgeted. That pricing strength tells us demand, not competitiveness, is the constraint. Services delivered exactly to plan (Rs 0.75 cr), confirming the recurring book is solid.
>
> "Gross margin of 28.5% is 150 bps below plan, split between input-cost inflation on goods and the revenue de-leverage. Opex ran Rs 5 lakh over on hires pulled forward from the Q4 plan. **Outlook:** the annual budget is back-loaded (Q4 goods Rs 2.85 cr on the March spike), so a 5% Q1 volume miss does not yet threaten the Rs 12 cr full-year number — *if* Q2 volume recovers. But two levers are now tight: the Rs 1.506 cr PBT floor has almost no headroom, and pulling opex forward has spent some of it early. Recommendation: hold the Q4 hires until Q2 volume confirms recovery, protecting ~Rs 5 lakh of PBT."

How it's reviewed

The CFO checks that the **bridge foots** — every piece sums to the Rs 15 lakh headline with no plug. She checks the **convention** (budget price on the volume leg, actual units on the price leg) so volume and price don't double-count. She checks that **favourable and unfavourable signs are right** (fewer units = unfavourable; higher price = favourable). And she reads the commentary for a *causal story with an action*, not a restatement of the table. The killer question: "so what do we do about it?"

Common mistakes & red flags

- **Reporting the net without the split.** "Goods down Rs 10.8 lakh" hides that volume (−15) and price (+4.2) are pulling in opposite directions — completely different management responses.
- **Double-counting the cross-term.** Using actual price on the volume leg AND actual units on the price leg over-attributes. Pick one convention and make the legs sum to the total.
- **Confusing margin-rate and margin-value.** A 150 bps rate drop and a revenue-driven GP fall are two different effects; separate them.
- **No outlook.** A variance with no forward read is a history lesson. Always answer "what does this mean for the full year."
- **Ignoring back-loading.** Judging the Rs 12 cr year off a soft Q1 when the budget itself put only 24% of goods in Q1 and 32% in Q4.

On the job & in the interview

The "why": variance analysis is where FP&A earns its seat — it converts a Rs 15 lakh gap into a decision (hold the hires, chase volume, defend price). The decomposition isolates *controllable* (price, opex) from *market* (volume) causes.

Jargon: **price-volume-mix (PVM)**, **favourable/unfavourable (F/U)**, **bridge / waterfall**, **rate vs volume effect**, **de-leverage**, **plug**.

Q: "Q1 revenue missed by Rs 15 lakh. Is that good or bad news, and why?" A: "Mixed, but structurally okay. The miss is entirely volume — 1,500 units at Rs 1,000, Rs 15 lakh — while price was actually *favourable*, ASP up Rs 20 for +Rs 4.2 lakh. Favourable price means we're not losing on competitiveness; we're demand-constrained. Given the budget is March-heavy, a 5% Q1 volume dip is recoverable, so I'd frame it as a demand watch-item, not a pricing crisis."

Q: "Walk me through a price-volume decomposition." A: "Revenue is units times price, so the variance splits two ways: volume variance is the unit change times *budget* price, and price variance is the price change times *actual* units. Holding one factor at budget and the other at actual makes the two legs sum exactly to the total with no residual. Here: volume (21,000−22,500)×1,000 = −15 lakh; price (1,020−1,000)×21,000 = +4.2 lakh; net −10.8 lakh on goods."

Q: "The margin also dropped to 28.5%. How do you separate that from the revenue miss?" A: "Two effects. The revenue shortfall costs GP at the budgeted 30% rate — 30% of the Rs 15 lakh miss is Rs 4.5 lakh. Separately, the *rate* fell 150 bps, which on Rs 2.70 cr of actual revenue is another Rs 4.05 lakh of GP. Splitting them tells management whether to chase volume or attack input costs — here, both."

The Monthly MIS / Management Pack

The ask

It's Friday, 10 July 2026, month-end +7 for June. The CFO sets the standing expectation:

> "Every month, by working-day 5 after close, I need the management pack — P&L actual vs budget vs last year, the KPI dashboard, the goods-vs-services split, and half a page of plain-English summary. It goes to the board and the bank. Same format every month, so I can read it in five minutes. And it can't take you five days to build — automate it."

Deadline: recurring, WD+5. Today's pack covers June / Q1 FY27. The job is a *repeatable* pack, not a one-off.

What you're given

The closed Tally GL for Q1 and the prior-year comparatives:

Q1 FY27 close (Rs cr):

Line	Actual	Budget	Prior Yr (Q1 FY26)
Revenue — goods	2.142	2.250	1.950
Revenue — services	0.750	0.750	0.620
Returns/adj	(0.192)	0.000	0.000
Total revenue	**2.700**	**2.850**	**2.570**
COGS	(1.931)	(1.995)	(1.850)
Gross profit	**0.769**	**0.855**	**0.720**
Employee cost	(0.290)	(0.270)	(0.245)
Other opex	(0.205)	(0.195)	(0.180)
EBITDA	**0.274**	**0.390**	**0.295**
Depreciation	(0.036)	(0.036)	(0.034)
Finance cost	(0.023)	(0.023)	(0.024)
PBT	**0.215**	**0.331**	**0.237**

KPI feeds: DSO 66 days (target 60); headcount 16 (budget 15); cash Rs 31 lakh.

Build it — step by step

Step 1 — One clean data feed. Export the Tally trial balance to a `GL_Dump` tab with columns: `GL code | GL name | Amount`. Map each GL to a P&L line via a lookup table (`Map` tab: `GL code → P&L line`). The pack's actual column is then one formula:

```
=SUMIFS(GL_Dump[Amount], GL_Dump[PLLine], $B5)
```

dragged down every P&L row. Budget and prior-year columns pull from their own stored tabs with the same key. **Nothing is typed twice** — next month you paste a fresh GL dump and the whole pack refreshes.

Step 2 — Variance columns. Two variance blocks: vs budget and vs prior year, each in value and %:

```
Var vs Bud (Rs)  = Actual − Budget
Var vs Bud (%)   = (Actual − Budget) / Budget     (guard with IFERROR for zero rows)
YoY growth (%)   = (Actual − PriorYr) / PriorYr
```

Step 3 — KPI tiles. A `Dashboard` tab computes the six headline KPIs with conditional-format traffic lights (green if within tolerance, red if not):

```
GM %       = GrossProfit / Revenue
EBITDA %   = EBITDA / Revenue
DSO        = Debtors / Revenue_annualised × 365
```

Step 4 — Segment view. A small goods-vs-services block so the two engines are always visible — this is where the mix story lives.

Step 5 — Chart choices. Keep it boring and legible: a **clustered column** for revenue (actual/budget/PY side by side), a **waterfall** for the PBT bridge, a **line** for DSO trend across months, and **KPI cards** for the six numbers. No pie charts, no 3-D. If the CFO reads it on a phone, it has to work in greyscale.

Step 6 — Power BI option (for the board version). The same model can live in Power BI off the Tally export. Core DAX measures:

```
Revenue     = SUM(GL[Amount])
GM %        = DIVIDE([Gross Profit], [Revenue])
Var vs Bud = [Revenue] − [Budget Amount]
YoY %       = DIVIDE([Revenue] − [Revenue PY], [Revenue PY])
```

and to pull prior year: `Revenue PY = CALCULATE([Revenue], SAMEPERIODLASTYEAR('Date'[Date]))`.

The deliverable

NTSPL Management Pack — Q1 FY2026-27

1. P&L — actual vs budget vs prior year (Rs cr)

Line	Actual	Budget	Var	Var %	PY	YoY %
Total revenue	2.700	2.850	(0.150)	−5.3%	2.570	+5.1%
Gross profit	0.769	0.855	(0.086)	−10.1%	0.720	+6.8%
GM %	28.5%	30.0%	−1.5pp		28.0%	+0.5pp
EBITDA	0.274	0.390	(0.116)	−29.7%	0.295	−7.1%
PBT	0.215	0.331	(0.116)	−35.0%	0.237	−9.3%

2. KPI dashboard

KPI	Actual	Target	Flag
Revenue (Rs cr)	2.70	2.85	🔴
Gross margin %	28.5%	30.0%	🔴
EBITDA (Rs cr)	0.274	0.390	🔴
DSO (days)	66	60	🔴
Headcount	16	15	🟡
Cash (Rs lakh)	31	35	🟡

3. Segment view (Rs cr)

Segment	Revenue	GM %	vs Budget
Goods	1.95 (net of adj)	25.0%	(0.30)
Services	0.75	45.0%	0.00
Blended	**2.70**	**28.5%**	**(0.15)**

4. Written summary (analyst voice):

> "June closes Q1 at Rs 2.70 cr revenue, 5% below budget but 5% ahead of last year — the miss is versus an ambitious plan, not a decline. The gap is goods volume (21,000 vs 22,500 units); pricing was favourable (ASP Rs 1,020). EBITDA at Rs 0.27 cr is the pressure point — the revenue miss de-leveraged margin (28.5% vs 30%) and opex ran Rs 4 lakh over on an early hire. DSO drifting to 66 days and cash at Rs 31 lakh (vs Rs 35 lakh opening) is a collections watch-item — Rs 6 lakh of the Rs 4 lakh cash drop is timing on receivables. **Actions:** (1) tighten collections to pull DSO back toward 60; (2) hold the remaining Q4 hires pending Q2 volume; (3) full-year Rs 12 cr still intact given the March-loaded plan, contingent on Q2 volume recovery."

How it's reviewed

The controller checks the pack **ties to the ledger** — total revenue and PBT in the pack must equal the Tally-closed numbers to the rupee. She checks **internal consistency**: EBITDA − depreciation − finance cost = PBT; segment revenue sums to total. She checks the **KPI flags** are computed, not hand-coloured. And the CFO checks the **summary is decision-oriented** and consistent with the variance pack from the prior chapter — the same Rs 15 lakh volume story, told the same way. Any number that appears in two places must match everywhere.

Common mistakes & red flags

- **Rebuilding from scratch monthly.** If the pack isn't a paste-and-refresh, WD+5 becomes WD+9. Invest once in the GL-map + SUMIFS engine.
- **Typing numbers into the output.** Every cell must trace to the GL dump or a stored budget/PY tab. Hand-keyed figures are how packs contradict the ledger.
- **No prior-year column.** Actual vs budget alone hides whether the business is growing. YoY reframes a "miss" (+5% YoY) honestly.
- **Vanity charts.** Pie/3-D charts that don't survive greyscale or a phone screen. Board packs are read fast — clustered columns and a waterfall.
- **KPIs with no target.** A DSO of 66 means nothing without the 60-day target and the flag. Always show the benchmark.
- **Inconsistent numbers across the pack.** Revenue Rs 2.70 cr on page 1 and Rs 2.69 cr on the chart destroys trust instantly.

On the job & in the interview

The "why": the MIS pack is FP&A's monthly heartbeat — it tells the board and the bank whether the plan is on track, in a format they can absorb in five minutes. Repeatability is the skill; a pack that takes a week to build is worthless by the time it's read.

Jargon: **MIS / management pack**, **actual vs budget vs PY**, **WD+5 (working-day 5)**, **tie-out to ledger**, **traffic-light KPIs**, **de-leverage**, **single source of truth**.

Q: "How do you build a monthly pack that's both fast and reliable?" A: "One clean data feed and zero hand-keying. I map every Tally GL to a P&L line once, then the actual column is a SUMIFS off the monthly GL dump; budget and prior-year sit in their own tabs on the same key. Next month I paste a fresh dump and the whole pack — P&L, KPIs, segment, charts — refreshes. It ties to the ledger by construction, and it's a WD+5 job, not a WD+9 one."

Q: "Why show prior year as well as budget?" A: "They answer different questions. Budget tells you if you hit the plan; prior year tells you if the business is actually growing. Our Q1 was 5% under budget but 5% up on last year — without the PY column you'd read that as pure bad news, when structurally the business is still expanding."

Q: "What KPIs would you put on the front page for a trading-and-services firm like this?" A: "Six: revenue, gross margin %, EBITDA, DSO, headcount, and cash. Revenue and GM% show the top line and mix health; EBITDA is the operating profit signal; DSO and cash flag the working-capital risk that a goods-heavy trader lives on; headcount tracks the biggest opex lever. Each with a target and a traffic light so the reader sees exceptions in two seconds."

The Rolling Forecast Update

The ask

It's **10 July 2026**. Q1 (Apr–Jun) actuals are closed and the variance pack you built last week is on the CFO's desk. In the Monday review she says:

> "The budget is now stale — we missed Q1 by Rs 15 lakh. I don't want to keep reporting against a number we already know is wrong. Give me a **full-year reforecast** by Friday: Q1 actual, plus a fresh view of Q2–Q4. I need to know where we now expect to *land* on revenue and PBT, and I need one page that explains **why the landing is different from budget** so I can walk the board through it. Keep the budget frozen as the yardstick — the reforecast is a separate column."

This is the **rolling forecast**: the budget stays fixed for the year (it's the promise), but every quarter you replace one more quarter of forecast with actuals and re-drive the rest. The deliverable is a **FY2026-27 reforecast that lands ~Rs 11.7 cr** on revenue, with a defendable driver story.

What you're given

Budget vs Q1 actual (the anchors):

Line	FY Budget	Q1 Budget	Q1 Actual	Q1 Var
Revenue	12.00	2.85	2.70	(0.15)
— Goods	9.00	2.25	2.10*	
— Services	3.00	0.60	0.60	on plan
Gross margin %	30.0%	30.0%	28.5%	(1.5pp)
Opex (emp+other+dep)	2.004	~0.50	0.55	(0.05)

*Rs cr. Goods Q1: 21,000 units vs 22,500 budget (volume miss), realised ASP Rs 1,020 vs Rs 1,000** (price gain partly offsetting).

Driver read from Q1: the volume shortfall is a **timing/demand** slip in the components line (two OEM orders pushed to Q2), not a lost account. Sales says the pipeline recovers but the full year won't fully catch up. Input costs (copper/PVC) ran hot, dragging gross margin. Two extra hires landed early (opex creep).

Build it — step by step

Rolling-forecast mechanics = Actual-to-date + Forecast-to-go (ATD + FTG). In Excel the full-year cell is never hard-typed — it's `=SUM(Q1_actual, Q2_fc, Q3_fc, Q4_fc)`. A single flag cell decides, per quarter, whether to pull actual or forecast:

```
FY_col  =SUMIFS(actuals,period,"≤"&last_closed) + SUMIFS(forecast,period,">"&last_closed)
```

Step 1 — Re-drive the goods volume. Budget was 90,000 units evenly (22,500/qtr). Reforecast: Q1 actual 21,000; the two pushed orders lift Q2, then a steady recovered run-rate:

Goods units	Q1	Q2	Q3	Q4	FY
Budget	22,500	22,500	22,500	22,500	90,000
Reforecast	21,000 (A)	21,300	21,500	21,500	**85,300**

Full-year units **85,300 vs 90,000 = −5,200 (−5.8%)** — we do *not* pretend Q1 fully recovers.

Step 2 — Hold the ASP gain. The Rs 1,020 realised price holds (a genuine list-price improvement, not a one-off). Goods reforecast = `85,300 × 1,020 = Rs 8.70 cr` (budget 9.00). Services stay on plan at **Rs 3.00 cr**.

`Revenue reforecast = 8.70 + 3.00 = Rs 11.70 cr` — the landing.

Step 3 — Re-drive margin. Q1 came in at 28.5%. Procurement has hedged part of the copper exposure, so H2 recovers toward budget but the full year is still below. Blend to **29.0%** → GP = `11.70 × 29.0% = Rs 3.393 cr`; COGS = Rs 8.31 cr.

Step 4 — Re-drive opex. The two early hires stick: employee cost `1.08 → 1.10`. Other opex, depreciation, finance cost held at budget (capex and the term loan are on schedule).

Step 5 — Roll to PBT/PAT.

```
EBIT = GP − Emp − Other − Dep = 3.393 − 1.10 − 0.78 − 0.144 = 1.369
PBT  = EBIT − Finance = 1.369 − 0.09 = 1.279
PAT  = PBT × (1 − 26%) ≈ 0.94   (25% + 4% cess)
```

The deliverable

FY2026-27 Reforecast (Rs cr) — budget frozen as yardstick:

Line	Budget	Q1 Actual	Q2–Q4 FTG	Reforecast	Var vs Bud	%
Revenue	12.00	2.70	9.00	**11.70**	(0.30)	−2.5%
COGS	(8.40)	(1.93)	(6.38)	**(8.31)**	0.09	
Gross profit	3.60	0.77	2.62	**3.39**	(0.21)	−5.8%
Gross margin %	30.0%	28.5%	29.2%	**29.0%**	(1.0pp)	
Employee	(1.08)	(0.28)	(0.82)	**(1.10)**	(0.02)	
Other opex	(0.78)	(0.20)	(0.58)	**(0.78)**	—	
Depreciation	(0.144)	(0.036)	(0.108)	**(0.144)**	—	
EBIT	1.596	0.254	1.112	**1.366**	(0.23)	−14.4%
Finance cost	(0.09)	(0.022)	(0.068)	**(0.09)**	—	
PBT	**1.506**	0.232	1.044	**1.276**	**(0.23)**	**−15.3%**
PAT (~26%)	1.11			**0.94**	(0.17)	−15%

Quarter phasing of the reforecast:

Rs cr	Q1 A	Q2 F	Q3 F	Q4 F	FY
Revenue	2.70	2.85	3.05	3.10	11.70
PBT	0.23	0.30	0.36	0.39	1.28

Analyst commentary (the one-pager): "We now expect to land the year at **Rs 11.70 cr revenue (−2.5% vs budget)** and **PBT Rs 1.28 cr (−15%)**. Revenue is off by only Rs 30 lakh — the goods volume miss (85,300 vs 90,000 units, the two Q2-slipped OEM orders never fully recovered) is *partly offset* by a genuine ASP improvement to Rs 1,020 and services holding exactly on plan. **The PBT fall is 6× the revenue fall** because two forces compound below the line: (i) input-cost-driven margin down 100bps, and (ii) two hires landing a quarter early. This is operating leverage in reverse — small revenue softness, amplified at the bottom. The recovery is real but back-ended: H2 PBT run-rate returns to plan."

How it's reviewed

The CFO's checks: - **ATD ties to the ledger.** Q1 reforecast column must equal the closed Q1 actuals to the rupee — no "smoothing" the past. - **Bridge closes.** Budget PBT 1.506 → reforecast 1.276. The Rs 23 lakh gap must decompose cleanly: revenue/volume, margin, opex. No plug. - **Drivers, not a haircut.** "We took 2.5% off revenue" is rejected; "85,300 units × Rs 1,020" is accepted. Every changed number traces to a driver with an owner (Sales owns volume, Procurement owns margin, HR owns headcount). - **Reforecast ≠ new budget.** The budget column is untouched; bonuses and board commitments still measure against it. - **Directional sense:** revenue −2.5% but PBT −15% must be *explained*, not buried.

Common mistakes & red flags

- **Re-opening the budget.** Overwriting the frozen budget destroys accountability. Keep budget, reforecast, and actual as three separate columns forever.

- **Hard-coding the FY total.** Full-year must be `=ATD + FTG` so it auto-updates when Q2 closes. A typed-over FY cell is the classic broken-model tell.
- **Sandbagging or hero numbers.** Assuming Q1's shortfall fully catches up (hero) or writing off the whole year (sandbag). Recovery must match the actual pipeline.
- **Changing volume *and* margin *and* opex with no owner.** If three drivers move, three names sign off.
- **Forgetting the price offset.** Netting volume and price into one "revenue miss" hides that pricing is actually helping — pros keep P and Q separate.
- **Margin creep from rounding.** Re-derive GP from COGS, don't apply a rounded %.

On the job & in the interview

The rolling forecast is the FP&A heartbeat — most Indian mid-market firms run a **quarterly reforecast** (larger ones monthly), always as ATD + FTG. The phrase to own: *"we replace a quarter of forecast with a quarter of actuals and re-drive the go-forward."*

Q: "Budget says 12, you're forecasting 11.7 — did you just cut the number?" "No — I re-drove it. Goods volume is 85,300 units vs 90,000 because two OEM orders slipped and don't fully recover; that's partly offset by ASP holding at Rs 1,020 and services on plan, landing revenue at Rs 11.70 cr. Every rupee of the Rs 30 lakh delta traces to a driver with an owner — it's a bottoms-up reforecast, not a top-down haircut."

Q: "Revenue is only down 2.5% but PBT is down 15% — why?" "Operating leverage in reverse. The Rs 21 lakh gross-profit loss combines the volume miss with a 100bps input-cost margin hit, and two hires landed a quarter early adding Rs 2 lakh of opex. Fixed opex doesn't flex down with revenue, so a small top-line slip amplifies at PBT. It also tells us the sensitivity: margin and headcount are the levers to defend, not the top line."

Q: "When do you reforecast vs stick with budget?" "Budget is the annual commitment and stays frozen for measurement. I reforecast whenever the actuals materially diverge from plan — here a −5% Q1 — so decision-makers steer off a current view. Two different jobs: budget for accountability, reforecast for steering."

Refreshing the 3-Statement Model (P&L to Balance Sheet to Cash Flow)

The ask

It's **16 July 2026**. The bank relationship manager has asked NTSPL for a **projected balance sheet and cash flow** to support the working-capital limit renewal, and the CFO wants the same model to answer her own question: "*if we hit the budget, what does the balance sheet and cash position look like at 31 Mar 2027, and can we service the term loan?*"

> "Refresh the three-statement model off the budget P&L. I want the P&L flowing into a **balance sheet** driven by our DSO/DPO/DIO, a proper **term-loan schedule**, and an **indirect cash flow** that reconciles to the closing cash. And it must **balance** — assets = liabilities + equity — with the check cell showing zero. Bank wants it Thursday."

The core skill: a **linked** model where P&L → Balance Sheet → Cash Flow, no hard-coded plugs, ending cash on the cash-flow statement equals cash on the balance sheet.

What you're given

Budget P&L (Rs cr, FY2026-27):

Line	Amount
Revenue	12.00
COGS	(8.40)
Gross profit	3.60
Employee + Other opex	(1.86)
Depreciation	(0.144)
EBIT	1.596
Finance cost	(0.09)
PBT	1.506
Tax (~26%)	(0.396)
PAT	**1.11**

Balance-sheet drivers (FY2026-27): DSO 60 days, DPO 45 days, DIO 40 days. Term loan **Rs 1.20 cr @ 9.5%**, repaid **Rs 15 lakh/quarter** (Rs 60 lakh/yr). Capex **Rs 40 lakh**; depreciation Rs 14.4 lakh. **Opening cash Rs 35 lakh.** No dividend.

Opening balance sheet (1 Apr 2026), Rs cr — the model's starting point:

Assets		Liab & Equity	
Net PPE	1.00	Creditors	0.90
Inventory	0.75	Term loan	1.20
Debtors	1.70	Share capital	0.50
Cash	0.35	Reserves	1.70
Total	**3.80**	**Total**	**3.80**

Build it — step by step

Step 1 — Working capital from ratios. Each closing balance is a driver × a P&L flow:

```
Debtors   = Revenue × DSO/365 = 12.00 × 60/365 = 1.97
Inventory = COGS    × DIO/365 = 8.40  × 40/365 = 0.92
Creditors = COGS    × DPO/365 = 8.40  × 45/365 = 1.04
```

These tie to the shared anchors (debtors ~2.0, inventory ~0.9, creditors ~1.05).

Step 2 — PPE roll-forward.

```
Closing PPE = Opening PPE + Capex − Depreciation = 1.00 + 0.40 − 0.144 = 1.256
```

Step 3 — Debt schedule (term loan). Interest is charged on the **average balance** so the P&L finance cost reconciles:

Term loan (Rs cr)	Q1	Q2	Q3	Q4	FY
Opening	1.20	1.05	0.90	0.75	1.20
Repayment	(0.15)	(0.15)	(0.15)	(0.15)	(0.60)
Closing	1.05	0.90	0.75	**0.60**	0.60
Interest @9.5% on avg	0.027	0.023	0.020	0.016	**~0.09**

Average FY balance ≈ Rs 0.90 cr; `0.90 × 9.5% ≈ Rs 0.086 cr ≈ Rs 0.09 cr` finance cost — matching the P&L anchor.

Step 4 — Equity roll-forward.

```
Closing reserves = Opening 1.70 + PAT 1.11 − Dividend 0 = 2.81
```

Step 5 — Indirect cash flow (rebuild cash bottom-up; it must land on the BS cash):

```
CFO = PAT + Depreciation − ΔDebtors − ΔInventory + ΔCreditors
    = 1.11 + 0.144 − (1.97−1.70) − (0.92−0.75) + (1.04−0.90)
    = 1.11 + 0.144 − 0.27 − 0.17 + 0.14 = 0.954
CFI = − Capex = −0.40
CFF = − Loan repayment = −0.60
Δ Cash = 0.954 − 0.40 − 0.60 = −0.046
Closing cash = 0.35 − 0.046 = 0.304  (≈ 0.30)
```

Step 6 — The circularity. Interest depends on the average debt balance; if the model funds shortfalls via a **revolver/overdraft**, interest also depends on the cash line — which depends on cash flow, which depends on interest. That's a **circular reference**. Two fixes: - **Enable iterative calculation:** *File → Options → Formulas → Enable iterative calculation, Max iterations 100, Max change 0.001.* Excel then solves the loop. - **Circularity switch (safer for auditability):** a flag cell `Circ_ON` (1/0). Interest = `IF(Circ_ON, rate×AVERAGE(open,close), rate×open)`. Turn it off to break `#REF!` /spiral errors, on to solve. Many shops charge interest on **opening balance only** to avoid the loop entirely — cleaner, marginally less precise.

The deliverable

Projected Balance Sheet, 31 Mar 2027 (Rs cr):

Assets		Liab & Equity	
Net PPE	1.256	Creditors	1.04
Inventory	0.92	Term loan	0.60
Debtors	1.97	Share capital	0.50
Cash	0.30	Reserves	2.81
Total assets	**4.45**	**Total L+E**	**4.45**
		Check (A − L−E)	**0.00**

Cash Flow Statement (indirect, Rs cr):

	FY
PAT	1.11
+ Depreciation	0.144
− Increase in debtors	(0.27)
− Increase in inventory	(0.17)
+ Increase in creditors	0.14
Cash from operations	**0.954**
Capex	(0.40)
Cash from investing	**(0.40)**
Loan repayment	(0.60)
Cash from financing	**(0.60)**
Net change in cash	**(0.046)**
Opening cash	0.35
Closing cash	**0.30**

Commentary: "On budget, NTSPL ends FY27 with a **balanced sheet at Rs 4.45 cr**, cash easing from Rs 35 to Rs 30 lakh. Operations throw off **Rs 95 lakh**, but Rs 30 lakh is absorbed by working-capital growth (debtors and inventory rising with the business), Rs 40 lakh goes to capex, and Rs 60 lakh de-levers the term loan to Rs 0.60 cr. The Rs 4.6 lakh cash draw is comfortable but thin — a bad debtors quarter would pressure it, which is exactly why the working-capital limit renewal matters. Debt service is easily covered: EBIT Rs 1.60 cr vs interest Rs 0.09 cr (interest cover ~18×)."

How it's reviewed

- **The check cell = 0.** Assets − (Liabilities + Equity) must be zero, not "Rs 400 rounding." A non-zero check = a broken link somewhere.
- **Cash ties twice.** Closing cash on the CF statement must equal cash on the balance sheet — the ultimate integrity test of a linked model.
- **No plugs.** Every BS line must roll from opening + a driver. If the controller finds a hard-typed number where a formula belongs, it fails.
- **Ratios back-check:** re-derive DSO from the model (`Debtors/Revenue×365 = 1.97/12×365 = 60`) — must return the input.
- **Interest reconciles** to the debt schedule, not a random 9.5% × 1.20.

Common mistakes & red flags

- **Plugging cash to force a balance.** The cardinal sin — if you type the cash figure to make A = L+E, the model is lying. Cash must *fall out* of the cash flow.
- **Sign errors on working capital.** An *increase* in debtors is a cash *outflow* (negative); an increase in creditors is an *inflow* (positive). Flip one sign and the check breaks by 2× the delta.

- **Double-counting depreciation.** It's added back in CFO (non-cash) *and* reduces PPE — both, not either.
- **Circular-reference spiral (`0` or `#REF!` everywhere).** Forgetting to enable iterative calc, or leaving it on when a genuine error enters. Keep the circularity switch.
- **Interest on closing (not average) balance** overstates the de-levering benefit; on opening overstates cost. Average is the honest middle — state which you use.
- **Tax on EBIT instead of PBT.** Tax is on profit *after* interest.

On the job & in the interview

The 3-statement model is the FP&A rite of passage — banks, boards, and diligence all want it. The mantra: **P&L drives the balance sheet, the balance sheet drives cash, and cash proves the P&L was real.**

Q: "Walk me through how a Rs 10 increase in depreciation flows through all three statements." "P&L: EBIT and PBT fall by 10, tax falls by ~2.6, so PAT falls by ~7.4. Cash flow: PAT down 7.4 but depreciation is added back +10, so CFO rises by +2.6 — the tax shield. Balance sheet: PPE falls 10, cash rises 2.6, and retained earnings fall 7.4; both sides move by −7.4 and it still balances."

Q: "What causes circularity and how do you handle it?" "Interest expense feeds the P&L, which drives cash, and if a revolver funds shortfalls, the debt balance depends on that cash — so interest depends on itself. I enable iterative calculation (100 iterations, 0.001 tolerance) with a circularity switch to break it if an error spirals, or I charge interest on the opening balance to avoid the loop entirely."

Q: "Your model won't balance — how do you find the error?" "Check the CF closing cash against BS cash first; the gap size often names the culprit. Then I audit working-capital signs, confirm every BS line rolls from opening + driver with no plug, and re-derive the ratios. Ninety percent of the time it's a sign flip on a working-capital movement or a missing add-back."

A CFO Decision-Support Ask: Quick DCF / NPV / IRR

The ask

It's **21 July 2026**, 4:40 pm. The CFO catches you before she leaves:

> "Sales wants to open a **Hyderabad-East branch** for the industrial-components line — new territory, dedicated stock, two field engineers. They're pitching it to the MD next week and they've handed me a spreadsheet that just says 'huge upside.' I need a **quick, honest read** by tomorrow noon: what's the **NPV** and **IRR** at our 12% cost of capital, and **how sensitive** is the answer to the discount rate and the growth assumption? One page — cash-flow table, the number, and your recommendation. If the terminal value is doing all the work, I want to know that."

This is classic **decision support**: a compact **DCF** on the initiative's free cash flow (FCFF), a terminal value, and `=NPV / =IRR` (and their dated cousins `=XNPV / =XIRR`) — plus a sensitivity grid so nobody hides behind a single point estimate.

What you're given

Initiative assumptions (FY2026-27 basis, Rs lakh):

Item	Value
Upfront investment (Year 0): fit-out + opening stock	40
WACC / discount rate	12%
Terminal growth (g)	3%
Forecast horizon	5 years
Tax	25% + cess (baked into FCFF)

Projected incremental Free Cash Flow to Firm (FCFF), Rs lakh — built by Sales from expected branch revenue less incremental COGS, opex, tax, and maintenance capex/working capital:

Year	1	2	3	4	5
FCFF	8	11	13	14	15

Build it — step by step

Step 1 — Lay out the cash-flow line. Row of years 0–5. Year 0 is the outflow `-40`; Years 1–5 are the FCFF above.

Step 2 — Terminal value (Gordon growth), computed at end of Year 5:

```
TV = FCFF₅ × (1 + g) / (WACC − g)
   = 15 × 1.03 / (0.12 − 0.03) = 15.45 / 0.09 = 171.67  (Rs lakh)
```

Step 3 — Discount factors `= 1/(1+WACC)^t` :

Year	1	2	3	4	5
DF @12%	0.8929	0.7972	0.7118	0.6355	0.5674

Step 4 — Present values:

Year	FCFF	+TV	Total CF	DF	PV
1	8		8	0.8929	7.14
2	11		11	0.7972	8.77
3	13		13	0.7118	9.25
4	14		14	0.6355	8.90
5	15	171.67	186.67	0.5674	105.93
				Σ PV	**139.99**

Of which PV of operating FCFF = 42.57 and **PV of terminal value = 97.42** — the terminal value is **~70% of enterprise value**. That's the flag the CFO asked for.

Step 5 — NPV and IRR (the Excel formulas):

```
Enterprise value  = SUM(PV) = 139.99
NPV  = 139.99 − 40 = 100.0    ' or directly:
NPV  = −40 + NPV(12%, C1:C5)   where C5 includes TV → ≈ 100
IRR  = IRR({−40, 8, 11, 13, 14, 186.67}) ≈ 51%
```

Note the `=NPV` **gotcha**: Excel's `NPV()` assumes the first cash flow is one period away, so you place the Year-0 outflow *outside* the function: `=-40 + NPV(12%, year1:year5)`. Never put Year 0 inside `NPV()`.

Step 6 — Dated version (`=XNPV / =XIRR` **)** for real, irregular cash-flow dates:

```
=XNPV(12%, values, dates)   ' values incl. −40 at the Year−0 date
=XIRR(values, dates)
```

`XNPV/XIRR` discount by actual calendar days, so a branch that opens mid-quarter is valued correctly — more precise than the annual `NPV/IRR`.

Step 7 — The terminal-value reality check. IRR of **51%** looks spectacular, but strip out the TV and the 5-year operating IRR on FCFF alone (`IRR{-40, 8, 11, 13, 14, 15}`) is only **~13.5%** — barely above the 12% hurdle. So the *headline return lives in the perpetuity assumption*, not the visible five years. That single sentence is the most valuable thing in the memo.

The deliverable

One-page decision memo — Hyderabad-East branch

Metric @ WACC 12%, g 3%	Value
PV of 5-yr FCFF	Rs 42.6 lakh
PV of terminal value	Rs 97.4 lakh
Enterprise value	Rs 140.0 lakh
Less: upfront investment	Rs (40.0) lakh
NPV	**Rs 100.0 lakh**
IRR (with TV)	**~51%**
IRR (operating flows only, ex-TV)	~13.5%
TV as % of EV	**~70%**

Sensitivity — NPV (Rs lakh) to WACC × terminal growth (2-var Data Table):

WACC \ g	2%	3%	4%
10%	123.8	142.1	166.5
12%	89.4	**100.0**	113.2
14%	66.6	73.3	81.4

Recommendation: "**Conditional GO.** At our 12% WACC the branch shows **NPV ~Rs 1.0 cr and IRR ~51%** — comfortably value-accretive, and it stays positive (Rs 67–166 lakh) across every WACC/growth combination we tested, so the *decision* is robust. **But** ~70% of the value sits in the terminal value, and the visible 5-year IRR is only ~13.5% — a whisker above the hurdle. My recommendation: approve, but (i) treat the first five years as the real underwriting case, (ii) sanity-check the FCFF ramp with Sales against the actual pipeline (these are their numbers, not audited), and (iii) build a stage-gate — release the Rs 40 lakh in two tranches, second tranche contingent on Year-1 FCFF ≥ Rs 6 lakh. The upside is real; the risk is that the perpetuity is doing the heavy lifting."

How it's reviewed

- **Year 0 outside `NPV()`.** The CFO will click the NPV cell — if `-40` sits inside the function, the answer is silently discounted an extra year and wrong.
- **TV timing.** Terminal value is a Year-5 figure, discounted by the Year-5 factor — not discounted twice, not left undiscounted.
- **TV as % of EV disclosed.** Any DCF where TV > 65% of value gets scrutiny; hiding it is a red flag.
- **IRR sign changes.** One sign flip (Year 0 negative, rest positive) → one IRR, reliable. Multiple flips can give multiple/► no IRR — then use NPV.
- **WACC consistency:** FCFF is a firm-level (unlevered) cash flow, so it's discounted at WACC — not cost of equity. Mixing them is an instant fail.

Common mistakes & red flags

- **Putting the initial outlay inside `=NPV()`.** The single most common Excel valuation bug — over-discounts everything.
- **Terminal value with `g ≥ WACC`.** The denominator `(WACC - g)` goes to zero or negative and TV explodes to infinity/nonsense. `g` must be modest (≤ long-run GDP, here 3%).
- **Trusting IRR alone.** IRR ignores scale and reinvestment; a 51% IRR on Rs 40 lakh may matter less than a 20% IRR on Rs 4 cr. Always pair IRR with NPV.
- **No sensitivity.** A single-point NPV invites a single-point argument. The Data Table shows the *range* and where it turns negative.
- **Terminal value doing all the work, undisclosed.** If the five visible years barely clear the hurdle, say so — don't let a perpetuity assumption smuggle the decision through.
- **Nominal vs real mismatch:** nominal cash flows need a nominal WACC and a nominal `g`. Don't mix.

On the job & in the interview

DCF/NPV/IRR is the language of capital allocation — every branch, machine, or acquisition passes through it. FP&A's value-add isn't the arithmetic (Excel does that); it's **stress-testing the assumptions** and telling the CFO *where the answer is fragile*.

Q: "NPV vs IRR — which do you trust and why?" "NPV, because it's in rupees and additive — it tells you how much value you create and you can sum projects. IRR is an intuitive % and great for ranking, but it breaks with unconventional cash flows (multiple sign changes give multiple IRRs), it's blind to scale, and it implicitly assumes reinvestment at the IRR itself. I lead with NPV and quote IRR alongside; if they disagree on a ranking, NPV wins."

Q: "Your NPV is Rs 1 crore — how confident are you?" "The *decision* is robust — NPV stays positive across every WACC (10–14%) and growth (2–4%) combination. The *magnitude* is not — about 70% of value is terminal, and the five-year operating IRR is only ~13.5%. So I'm confident it's value-accretive but I'd underwrite it on the visible cash flows and stage-gate the funding, rather than bank on the perpetuity."

Q: "Why discount FCFF at WACC and not the cost of equity?" "FCFF is the cash available to *all* capital providers — debt and equity — before financing flows, so it must be discounted at the blended cost of both, the WACC (~12% here). Cost of equity pairs with FCFE, the cash to equity holders only. Match the cash flow to the discount rate or you double-count or miss the financing effect."

Scenario & Sensitivity for a Board Question

The ask

It's **24 July 2026**. The quarterly board meeting is in five days. The CFO forwards you the Chair's email:

> "Given the Q1 volume miss and the noise on copper prices, the board wants to understand our **downside**. Specifically: *what happens to profit if volumes fall 10% and input costs rise 5%?* Show me a **best / base / worst** view, and I want to see **which one variable moves profit the most** so we know where to focus. Keep it to **one slide** — the board doesn't read models, they read the punchline."

Your job: build a **scenario switch** (best/base/worst on one toggle), Excel **What-If Data Tables** (1-variable and 2-variable) to map the sensitivity, a **tornado chart** ranking the drivers by their swing on PBT, and distil it to one slide. Base case = the FY2026-27 budget (Revenue Rs 12.00 cr, PBT Rs 1.506 cr).

What you're given

Base case = budget, with the drivers exposed as switchable inputs (Rs cr unless noted):

Driver	Base value
Goods volume	90,000 units
Goods ASP	Rs 1,000
Goods COGS / unit	Rs 750 (25% margin)
Services revenue	3.00
Services COGS	1.65 (55% of rev)
Fixed opex (emp + other + dep)	2.004
Finance cost	0.09

Base P&L (checks to anchors): Revenue 12.00, COGS 8.40, GP 3.60, EBIT 1.596, **PBT 1.506**.

Scenario definitions (agreed with the CFO):

Lever	Worst	Base	Best
Goods volume	−10%	0%	+5%
Input cost (all COGS)	+5%	0%	−2%

Build it — step by step

Step 1 — The scenario switch. One cell `$B$1 = scenario number` (1 worst / 2 base / 3 best). Each driver reads its value from a small lookup using `CHOOSE` (or `INDEX`):

```
Volume_used  = CHOOSE($B$1, 81000, 90000, 94500)
Cost_factor  = CHOOSE($B$1, 1.05, 1.00, 0.98)
' INDEX equivalent, cleaner for many drivers:
Volume_used  = INDEX(Volume_row, $B$1)
```

Flip `$B$1` and the whole P&L re-solves. This is the **live scenario toggle** the CFO clicks in the meeting.

Step 2 — Wire the P&L to the switched drivers:

```
Goods rev   = Volume_used × 1000 / 1e7           (Rs cr)
Goods COGS  = Volume_used × 750 × Cost_factor /1e7
Svc COGS    = 1.65 × Cost_factor
Revenue     = Goods rev + 3.00
COGS        = Goods COGS + Svc COGS
PBT         = Revenue − COGS − 2.004 − 0.09
```

Step 3 — Compute the three scenarios.

Worst: units 81,000 → goods rev `8.10`; goods COGS `81,000×750×1.05 = 6.38`; svc COGS `1.65×1.05 = 1.73`; COGS `8.11`; GP `2.99`; PBT `2.99 − 2.094 = 0.90`. *Best:* units 94,500 → goods rev `9.45`; goods COGS `94,500×750×0.98 = 6.95`; svc COGS `1.617`; COGS `8.56`; GP `3.89`; PBT `1.79`.

Step 4 — 1-variable Data Table (PBT vs goods volume). Column of unit values, one formula cell `=PBT`, then *Data → What-If Analysis → Data Table*, column input cell = the volume driver:

Goods units	81,000	85,500	90,000	94,500	99,000
PBT (Rs cr)	1.28	1.39	**1.51**	1.62	1.73

(Volume alone, input cost held at base.)

Step 5 — 2-variable Data Table (the board's exact question). Rows = volume change, columns = input-cost change; top-left corner cell = `=PBT`; row input = cost factor, column input = volume:

PBT (Rs cr) — vol ↓ / cost →	+5%	0%	−5%
−10%	**0.90**	1.28	1.67
−5%	0.99	1.39	1.80
0%	1.09	**1.51**	1.93
+5%	1.18	1.62	2.06

The board's scenario (volume −10%, cost +5%) is the **top-left corner: PBT Rs 0.90 cr** — a 40% fall, but still solidly profitable.

Step 6 — Tornado (drivers ranked by PBT swing). Flex each driver one at a time by a comparable range and record the PBT swing (high − low):

Driver	Flex	PBT low	PBT high	Swing
Goods ASP	±5%	1.06	1.96	**0.90**
Input cost (COGS)	±5%	1.09	1.93	**0.84**
Goods volume	±10%	1.28	1.73	**0.45**
Services revenue	±10%	1.37	1.64	**0.27**
Fixed opex	±5%	1.41	1.61	**0.20**

Sorted widest-to-narrowest, this is the tornado: **price and input cost dominate**; volume is third; opex barely moves the needle.

The deliverable

One slide — "Downside sensitivity of FY27 PBT"

Scenario summary (Rs cr):

	Worst	Base	Best
Revenue	11.10	12.00	12.45
Gross profit	2.99	3.60	3.89
EBIT	0.98	1.60	1.88
PBT	**0.90**	**1.51**	**1.79**
vs base	−40%	—	+19%

The board's question, answered: "*If volumes fall 10% and input costs rise 5% simultaneously, PBT lands at Rs 0.90 cr — down 40%, but still firmly positive. NTSPL does not go loss-making even in the compound-downside corner.*"

Where profit is most sensitive (tornado punchline): "*PBT is most exposed to selling price (±Rs 0.90 cr) and input cost (±Rs 0.84 cr) — the margin levers — far more than to volume (±Rs 0.45 cr). Management focus should be a pricing pass-through clause on AMC/goods and copper hedging, not chasing volume. A 5% price increase alone offsets the entire modelled cost shock.*"

Analyst commentary: "The business is margin-sensitive, not volume-sensitive — a 5% move in price or input cost swings PBT twice as hard as a 10% volume move, because volume drags COGS with it while price and input cost fall straight to the bottom line. The downside is uncomfortable but survivable; the *actionable* message is to defend margin, and the cheapest insurance is a contractual price-escalation on new orders."

How it's reviewed

- **Base scenario ties to budget.** With `$B$1 = 2` the model must reproduce PBT Rs 1.506 cr exactly — if the switch drifts off budget, every scenario is wrong.
- **Corner check.** The 2-var table's worst corner must equal the standalone worst scenario (Rs 0.90 cr both ways).

- **Data Table input cells correct.** Row/column input cells must point at the actual driver cells, not the formula — the classic Data Table mis-wire produces a flat grid.
- **Tornado symmetry sanity.** Linear drivers give near-symmetric swings; a wildly asymmetric bar signals a formula error or a genuine non-linearity worth flagging.
- **One decision, not ten numbers.** The CFO wants the punchline (still profitable; margin is the lever) — the grid is evidence, not the message.

Common mistakes & red flags

- **Hard-coding scenario outputs** instead of a live switch. When the board asks "what if volume falls 8%, not 10%?" a hard-coded deck can't answer; a `CHOOSE` / `INDEX` -driven model re-solves in one keystroke.
- **Data Table input cell pointing at the wrong reference** → the whole table returns the base value in every cell. Always test one corner by hand.
- **Flexing everything at once and calling it sensitivity.** Sensitivity = one variable at a time (that's the tornado); scenarios = a *coherent* combination. Don't confuse them.
- **Volatile Data Tables slowing the file.** Large 2-var tables recalc on every edit — set *Calculation → Automatic except for Data Tables* on big models.
- **Ranking the tornado by raw driver size, not PBT impact.** A big revenue line with thin margin may swing PBT less than a small high-margin one. Rank by the *output* swing.
- **Presenting worst case with no probability or action.** "PBT could be Rs 0.90 cr" is scary and useless alone; "...and here's the pricing/hedging lever that protects it" is decision-grade.

On the job & in the interview

Scenario and sensitivity work is where FP&A earns its seat: turning "what if?" into a decision. Boards think in ranges and downside, not point estimates — the analyst who can toggle a live model in the room is worth ten static decks.

Q: "Difference between a scenario and a sensitivity?" "A sensitivity flexes *one* variable and holds the rest — it isolates which lever matters most (that's the tornado). A scenario moves a *consistent set* of variables together to tell a coherent story — 'demand recession' might combine lower volume, lower price, and higher cost at once. Sensitivity finds the levers; scenarios tell the story. I use both: tornado to prioritise, best/base/worst to communicate."

Q: "How do you build a scenario switch in Excel?" "A single selector cell drives every input via `CHOOSE(selector, worst, base, best)` or `INDEX(range, selector)` . Each driver has its three values in a table and reads the chosen column. Flip the one cell and the whole model re-solves — no copy-paste, fully auditable, and I can demo live in the board meeting. For decision variables I'll layer a two-variable Data Table on top for the grid."

Q: "The board asks 'what's our biggest risk to profit?' — how do you answer with data?" "I'd point to the tornado: PBT swings most on selling price (±Rs 0.90 cr) and input cost (±Rs 0.84 cr), roughly double the volume sensitivity, because those hit margin directly. So the biggest risk isn't losing units — it's margin compression from copper prices without a pricing pass-through. That reframes the discussion from 'sell more' to 'protect the spread,' and I'd recommend a price-escalation clause and a hedging policy as the mitigations."

Unit Economics & KPI Deep-Dive

The ask

It's **20 July 2026**. The CFO forwards you an email chain with the Managing Director. The MD has been reading about "unit economics" and SaaS-style LTV/CAC metrics, and he wants to know, in his words: "*For every box we sell and every AMC we sign, what do we actually make — and how many do we need to sell before we stop losing money? Give me the per-unit picture and the two or three numbers I should watch every month.*"

The CFO wants a one-pager by **Thursday 23 July**: goods contribution per unit, services contribution per contract, the company break-even, a customer-acquisition (CAC) vs lifetime-value (LTV) read on the AMC book, and a **KPI tree** that shows how these roll up to the Rs 12.00 cr revenue and Rs 1.11 cr PAT already in the FY2026-27 budget. Each KPI must name the action it drives — no vanity metrics.

What you're given

The budget anchors, pulled from the approved FY2026-27 plan:

Line	Goods	Services	Total
Revenue	9.00 cr	3.00 cr	12.00 cr
Volume driver	90,000 units	250 AMC contracts	—
Unit price	ASP Rs 1,000	Rs 1,20,000 / contract	—
Gross margin %	25%	45%	30%
Gross profit	2.25 cr	1.35 cr	3.60 cr

Fixed cost base (below gross profit) from the budget:

Fixed / period cost	FY2026-27
Employee cost	Rs 1.08 cr
Other opex	Rs 0.78 cr
Depreciation	Rs 0.144 cr
Finance cost	Rs 0.09 cr
Total fixed	**Rs 2.094 cr**

AMC book facts from the CRM (Tally + a HubSpot export): 250 live contracts, average tenure ~4 years (retention ~75%/yr, so churn ~25%), and the sales/marketing spend attributable to *winning new AMC customers* this year is about **Rs 18 lakh** for **~60 new logos**.

Build it — step by step

Step 1 — Goods contribution per unit. ASP is Rs 1,000, gross margin 25%, so unit cost = Rs 750 and **contribution per unit = Rs 250** (this equals gross profit per unit because there's no separately variable selling cost per box). In Excel I lay it as a driver block:

```
ASP            =1000
GM%_goods      =0.25
Unit_cost      =ASP*(1-GM%_goods)       → 750
Contrib_unit   =ASP-Unit_cost           → 250
Contrib_margin =Contrib_unit/ASP        → 25%
```

Total goods contribution = 90,000 × 250 = **Rs 2.25 cr** — ties to budget gross profit for goods.

Step 2 — Services contribution per contract. Price Rs 1,20,000, margin 45%, so variable delivery cost = Rs 66,000 and **contribution per contract = Rs 54,000**. Across 250 contracts = Rs 1.35 cr — ties out.

Step 3 — Blended contribution & break-even. The company sells a mix, so I compute a **weighted contribution margin**, not a simple average:

```
Blended CM% = Total contribution / Total revenue
            = 3.60 cr / 12.00 cr = 30%
```

Break-even revenue = Fixed cost ÷ blended CM%:

```
BE_revenue = 2.094 / 0.30 = Rs 6.98 cr
```

To express it in **goods-equivalent units** if the whole gap were closed with boxes, contribution needed above services = fixed cost less services contribution:

```
Fixed to cover by goods = 2.094 - 1.35 = 0.744 cr
BE goods units          = 74,40,000 / 250 = 29,760 units
```

So at the current AMC book, NTSPL breaks even at **~29,760 boxes/year** (~33% of the 90,000 plan) — a comfortable margin of safety.

Margin of safety = (12.00 − 6.98) / 12.00 = **41.8%**. Revenue can fall 41.8% before NTSPL hits break-even.

Step 4 — CAC and LTV on the AMC book.

```
CAC = New-AMC S&M spend / New logos = 18,00,000 / 60 = Rs 30,000 per customer
Annual contribution/contract        = Rs 54,000
Avg lifetime (yrs) = 1/churn = 1/0.25 = 4 years
LTV (undiscounted)  = 54,000 x 4      = Rs 2,16,000
LTV:CAC             = 2,16,000 / 30,000 = 7.2x
CAC payback (months)= 30,000 / (54,000/12) = 6.7 months
```

A LTV:CAC of **7.2x** and a **~7-month** payback is healthy (the rule of thumb is LTV:CAC ≥ 3x and payback < 12 months). If anything it signals NTSPL is *under-investing* in AMC acquisition — there's room to spend more to win logos.

Step 5 — the KPI tree. I model revenue as `Volume × Price × Margin`, so each leaf is an operational lever a manager can move.

The deliverable

Unit economics summary — FY2026-27

Metric	Goods	Services
Price (ASP)	Rs 1,000/unit	Rs 1,20,000/contract
Variable cost	Rs 750	Rs 66,000
Contribution/unit	**Rs 250**	**Rs 54,000**
Contribution margin	25%	45%
Annual volume	90,000	250
Total contribution	Rs 2.25 cr	Rs 1.35 cr

Company economics: blended CM 30% · fixed cost Rs 2.094 cr · **break-even Rs 6.98 cr** (~29,760 goods units) · margin of safety **41.8%** · AMC **LTV:CAC 7.2x**, payback 6.7 months.

KPI tree (how the numbers roll up):

```
                    PAT Rs 1.11 cr
                          |
           EBIT Rs 1.596 cr  =  Contribution 3.60cr − Fixed 2.094cr(− Fin cost in PBT)
          /                              \
 Goods contribution 2.25cr        Services contribution 1.35cr
      /        |        \                 /        |        \
Units    ASP Rs1,000  CM% 25%      Contracts  Price Rs1.2L  CM% 45%
90,000                              250
   |          |          |            |          |           |
demand/    pricing    input cost   retention   pricing     delivery
coverage   discipline  & mix       + new logos             efficiency
```

Analyst commentary: *Goods carry the volume but the thin 25% margin means price discipline and input cost are the swing factors — a Rs 20 ASP slip on 90,000 units is Rs 18 lakh of contribution gone. Services are the profit engine per rupee (45% CM, 7.2x LTV:CAC); with a sub-7-month payback we are arguably starving AMC growth. Recommend the MD treat AMC net-adds and goods ASP realisation as the two board-level KPIs.*

KPI → action map:

KPI	Owner	Drives the action
Goods ASP realisation (actual vs Rs 1,000)	Sales	Tighten discount approvals; protect contribution
Goods units vs plan	Sales	Pipeline coverage, channel expansion
Gross margin % (goods)	Procurement	Renegotiate input cost, manage product mix
AMC net-adds (new − churned)	Services	Fund acquisition (payback only 6.7 mo)
AMC retention %	Services	Renewal calls, SLA quality → protect LTV
CAC & LTV:CAC	FP&A	Set the S&M budget ceiling

How it's reviewed

The CFO's checks: (1) **tie-out** — goods contribution Rs 2.25 cr + services Rs 1.35 cr = Rs 3.60 cr = budget gross profit; break-even and unit numbers must reconcile to the approved plan, not float free. (2) **Blended vs simple margin** — she'll confirm you weighted the CM by revenue mix (30%), not averaged 25% and 45% to a wrong 35%. (3) **CAC definition** — is the Rs 18 lakh *only* new-logo acquisition spend, or does it wrongly include renewal/delivery cost? (4) **LTV conservatism** — did you use churn-implied lifetime and flag that it's undiscounted? (5) Every KPI has a named owner and action.

Common mistakes & red flags

- **Averaging the two margins** to 35% instead of revenue-weighting to 30% — overstates profitability and understates break-even.
- **Treating depreciation/finance cost as variable** — they're fixed; putting them in the per-unit cost double-counts and distorts contribution.
- **Vanity LTV** — using revenue (Rs 1.2 L) not contribution (Rs 54k) in LTV inflates it ~2.2x. LTV must be a *margin* figure.
- **Ignoring churn** in lifetime — assuming contracts last forever gives an infinite LTV.
- **Confusing gross margin with contribution margin** — here they coincide for goods, but the moment a per-unit selling/logistics cost appears they diverge; keep the definitions explicit.
- **KPIs with no action** — "revenue" alone isn't a KPI you can act on; decompose to volume/price/margin so someone can *do* something.

On the job & in the interview

Unit economics is how FP&A turns a P&L into levers. The CFO doesn't act on "gross profit fell"; she acts on "ASP slipped Rs 20 and units missed 1,500." The **contribution margin** (price − variable cost) is the number that scales with volume; **fixed cost ÷ CM% = break-even**; **margin of safety** tells you how much cushion you have; **LTV:CAC and payback** tell you whether growth spend pays.

Q: "We sell goods at 25% margin and services at 45%. What's the blended margin and why can't I just average them?" *A: You weight by revenue, not count. 9cr×25% + 3cr×45% = 2.25 + 1.35 = 3.60cr on 12cr = 30%. A simple average (35%) assumes an equal revenue split, which we don't have — goods are 75% of revenue and drag the blend down toward 25%.*

Q: "Our AMC LTV:CAC is 7x. Good news?" *A: It's healthy — above the 3x bar with a ~7-month payback. But a very high ratio often means we're under-spending on acquisition. Given the short payback, I'd model spending more to win logos: as long as incremental payback stays under ~12 months and we can deliver the SLA, growth is accretive. I'd watch retention closely because LTV is only as good as the churn assumption.*

Q: "How many boxes to break even?" *A: Fixed cost Rs 2.094cr, blended CM 30% → break-even revenue Rs 6.98cr. Holding the 250 AMC contracts, goods must cover Rs 0.744cr of fixed cost at Rs 250/unit contribution = ~29,760 units, about a third of our 90,000 plan. Margin of safety is ~42%.*

A Capex Business Case

The ask

It's **10 August 2026**. The COO wants to buy a **warehouse-automation system** — conveyors, barcode scanning, a small WMS integration — to speed up despatch and cut manual handling. Quoted cost **Rs 40 lakh**, inside the FY2026-27 Rs 40 lakh capex budget. He claims it saves **Rs 12 lakh a year** in labour, error and rework over a **6-year** life, with a **Rs 4 lakh** salvage at the end.

The CFO drops it on your desk: "*Before this goes to the board on the 20th, run the appraisal. Payback, NPV, IRR at our 12% hurdle, and a sensitivity. One page, with a go/no-go and the risks. Don't just trust the Rs 12 lakh — tell me what happens if it's wrong.*"

What you're given

Input (FY2026-27)	Value
Initial outlay (Year 0)	Rs 40,00,000
Annual saving (pre-tax cash)	Rs 12,00,000
Life	6 years
Salvage (end of Year 6)	Rs 4,00,000
Discount rate (WACC)	12%

For this appraisal the CFO says to keep it on a **pre-tax cash basis** (the board sees pre-tax project economics; tax shields are a second-order refinement noted separately). Cash flows are treated as arising at year-end.

Build it — step by step

Step 1 — lay out the cash-flow schedule. In Excel, one column per year, Year 0 negative:

Year	Cash flow	Formula
0	−40,00,000	outlay
1	12,00,000	saving
2	12,00,000	saving
3	12,00,000	saving
4	12,00,000	saving
5	12,00,000	saving
6	16,00,000	12,00,000 saving + 4,00,000 salvage

Step 2 — payback (simple). Cumulative cash recovers the Rs 40 lakh outlay:

```
End Y1: -28,00,000   End Y2: -16,00,000   End Y3: -4,00,000   End Y4: +8,00,000
Payback = 3 + 4,00,000/12,00,000 = 3.33 years  (~3 yrs 4 months)
```

Step 3 — discounted payback. Discount each year at 12% (factor = 1/1.12^n), then re-cumulate:

Year	CF	DF @12%	PV	Cum PV
1	12,00,000	0.8929	10,71,429	−29,28,571
2	12,00,000	0.7972	9,56,633	−19,71,939
3	12,00,000	0.7118	8,54,136	−11,17,802
4	12,00,000	0.6355	7,62,621	−3,55,182
5	12,00,000	0.5674	6,80,912	+3,25,730
6	16,00,000	0.5066	8,10,634	+11,36,364

Discounted payback = 4 + 3,55,182/6,80,912 = **4.52 years** (~4 yrs 6 months).

Step 4 — NPV. Sum of PVs of Years 1–6 less the outlay. In Excel:

```
=NPV(12%, 12L,12L,12L,12L,12L,16L) − 40L
```

Sum of PV inflows = 10,71,429 + 9,56,633 + 8,54,136 + 7,62,621 + 6,80,912 + 8,10,634 = **Rs 51,36,364**.

```
NPV = 51,36,364 − 40,00,000 = Rs 11,36,364  (positive)
```

Step 5 — IRR. The rate where NPV = 0. In Excel, list CFs Year 0–6 and `=IRR(range)`. Solving:

```
At 12%: NPV = +11.36L ;  At 25%: NPV = −1.7L (approx)
IRR ≈ 23.4%
```

Well above the 12% hurdle. `=IRR({-40L,12L,12L,12L,12L,12L,16L})` returns **~23.4%**.

Step 6 — sensitivity. The soft number is the Rs 12 lakh saving, so I flex it and the discount rate. NPV (Rs lakh):

Annual saving →	Rs 9 L	Rs 12 L	Rs 15 L
Discount 10%	+2.6	+16.2	+29.8
Discount 12%	−1.2	+11.4	+23.9
Discount 14%	−4.6	+7.0	+18.7

Break-even saving (NPV = 0 at 12%): the project needs roughly **Rs 9.7 lakh/yr** to clear the hurdle — a ~19% cushion below the Rs 12 lakh claim. Below that, or if the saving falls to Rs 9 L, the case turns negative.

The deliverable

CAPEX MEMO — Warehouse Automation System To: CFO / Board · **From:** FP&A · **Date:** 12 August 2026

Recommendation: GO. The Rs 40 lakh warehouse-automation investment clears every test at the 12% hurdle.

Metric	Result	Hurdle	Verdict
NPV @12%	**+Rs 11.36 lakh**	> 0	Pass
IRR	**~23.4%**	> 12%	Pass
Simple payback	3.33 yrs	< 4 yrs	Pass
Discounted payback	4.52 yrs	< 6-yr life	Pass
Break-even saving	~Rs 9.7 L/yr	vs Rs 12 L claimed	19% cushion

Commentary: The project returns 23.4% against our 12% cost of capital and adds Rs 11.4 lakh of value in today's money. It recovers cash in about 3 years 4 months (4.5 on a discounted basis, comfortably inside the 6-year life). The economics only turn negative if annual savings fall below ~Rs 9.7 lakh — a 19% shortfall — so there is a reasonable buffer against optimism in the labour-saving estimate.

Risks & mitigants: - **Savings realisation** — Rs 12 L assumes headcount is genuinely redeployed/not backfilled. Mitigant: tie sign-off to a post-implementation labour-cost review at Year 1. - **Implementation slip** — WMS integration with Tally could overrun; delayed go-live pushes Year-1 saving out and cuts NPV. Mitigant: fixed-price vendor SOW, hold 10% retention. - **Volume dependency** — savings scale with despatch volume; a demand downturn (see Q1 volume miss) shrinks the benefit. Mitigant: the ~19% break-even cushion absorbs moderate slippage. - **Salvage uncertainty** — Rs 4 L in Year 6 is small (PV ~Rs 2 L); immaterial to the decision.

Ask: approve Rs 40 lakh capex; release against a fixed-price SOW with a Year-1 savings audit.

How it's reviewed

The CFO checks: (1) **sign convention** — outlay negative in Year 0, and NPV subtracts the outlay (a classic `=NPV()` error is including Year 0 *inside* the function, which wrongly discounts it one period). (2) **Salvage in the right year** — Rs 4 L sits in Year 6, added to that year's saving. (3) **IRR sanity** — 23.4% vs 12% hurdle, and that IRR and NPV agree in direction (both say go). (4) **Sensitivity honesty** — did you flex the *right* driver (the saving), and show the break-even? (5) **Hurdle rate** — 12% WACC, consistent with the appraisals elsewhere in the plan. (6) Payback used as a liquidity check, not the decision — NPV rules.

Common mistakes & red flags

- **Including Year 0 inside `=NPV()`** — Excel's NPV assumes the first value is one period out. Correct form: `=NPV(rate, Y1:Y6) + Year0` (with Year 0 negative), or subtract the outlay outside.
- **Forgetting salvage** or putting it in the wrong period.
- **Using payback to decide** — payback ignores time value and everything after the payback year; it's a screen, not the verdict.

- **No sensitivity** — a single-point NPV hides that the case rests on an unaudited Rs 12 L claim. Always show break-even.
- **Double-counting depreciation as a cash outflow** — on a pre-tax cash basis depreciation is non-cash; the Rs 40 L is captured once, at Year 0. (If you build the after-tax version, depreciation re-enters only as a *tax shield*.)
- **Mismatched rate** — appraising at cost of debt (9.5%) instead of WACC (12%) flatters the NPV.

On the job & in the interview

Capex appraisal is where FP&A protects the balance sheet. The board wants three things: does it create value (**NPV > 0**), does it beat our cost of capital (**IRR > WACC**), and how fast do we get our cash back (**payback**). NPV is the decision metric because it's in rupees and additive; IRR is the intuitive rate; payback is the liquidity comfort. The real skill is **sensitivity** — pressure-testing the one soft assumption.

Q: "NPV says Rs 11 lakh, IRR says 23%. Which do you trust and why?" *A: NPV for the decision — it's an absolute rupee value of wealth created and it's additive across projects. IRR is a useful communication number (23% vs a 12% cost of capital), but it can mislead with non-conventional cash flows or when comparing projects of different scale. Here they agree, so it's an easy go; when they conflict, I follow NPV.*

Q: "The COO's Rs 12 lakh saving — how do you protect the company if he's optimistic?" *A: I ran the break-even: the project still clears 12% down to about Rs 9.7 lakh a year, a 19% cushion. Below that it destroys value. So I'd recommend approval but tie the release to a fixed-price SOW and a Year-1 labour-cost audit — if the redeployment doesn't happen, we catch it before Year 2 and the salvage/exit is cheap.*

Q: "Why 12% and not our 9.5% loan rate?" *A: We fund with a mix of debt and equity; equity holders demand more than 9.5%. The blended WACC (~12%) is the true opportunity cost of the capital tied up, so it's the right hurdle. Discounting at the debt rate alone would overstate NPV and wave through projects that don't actually cover the cost of the equity funding them.*

Working Capital & the 13-Week Cash Forecast

The ask

It's **Monday, 6 July 2026**. The term-loan sanction letter carries a covenant: **minimum cash balance of Rs 20 lakh** at all times, tested on the bank's month-end certificate. With the Q1 revenue miss (Rs 2.70 cr actual vs Rs 2.85 cr budget) and a big supplier payment and an advance-tax instalment both due this quarter, the CFO is nervous about a mid-quarter dip.

She asks for two things by **Wednesday**: (1) a clean read on **working capital and the cash conversion cycle** — how long NTSPL's cash is tied up — and (2) a **13-week direct cash-flow forecast** (a rolling treasury tool) showing every week's collections, payments, payroll, GST, TDS and loan EMI, with the **minimum-cash covenant check** flagged and a list of **levers to free cash** if a week breaches.

What you're given

Working-capital assumptions and balances (FY2026-27):

Item	Assumption	Balance
Debtors (DSO)	60 days	Rs 2.00 cr
Inventory (DIO)	40 days	Rs 0.90 cr
Creditors (DPO)	45 days	Rs 1.05 cr
Term loan	Rs 1.20 cr @ 9.5%	—
Opening cash (Mon 6 Jul)	—	Rs 35 lakh

Flow anchors (annualised): revenue Rs 12.00 cr (~Rs 1.00 cr/month, ~Rs 23 lakh/week), COGS Rs 8.40 cr, monthly payroll ~Rs 9 lakh (employee cost 1.08 cr/12). GST is broadly net ~Rs 3–4 lakh/month payable (output less input credit), TDS deposits ~Rs 1.5 lakh/month, advance tax Q1 instalment (15%) due 15 September but the Rs 15 lakh supplier catch-up lands in Week 4.

Build it — step by step

Step 1 — the cash conversion cycle (CCC). This is how many days cash is locked in operations:

```
CCC = DSO + DIO − DPO
    = 60  + 40  − 45  = 55 days
```

NTSPL waits **55 days** between paying for goods/inventory and collecting the cash from customers. At ~Rs 12 cr revenue, each day of CCC ties up roughly `12cr/365 ≈ Rs 3.29 lakh`. So the 55-day cycle locks up about **Rs 1.81 cr** of operating cash — money funded by the term loan and cash reserves. Cutting CCC by 10 days frees ~Rs 33 lakh.

Cross-check via balances: Debtors 2.00 + Inventory 0.90 − Creditors 1.05 = **Net working capital Rs 1.85 cr** — consistent with the 55-day read.

Step 2 — frame the 13-week grid. In Excel, weeks as columns (W1 = w/c 6 Jul … W13 = w/c 28 Sep), rows as line items. The engine:

```
Closing cash (Wn) = Opening cash (Wn) + Collections − all Payments
Opening cash (Wn+1) = Closing cash (Wn)          [carry forward]
Covenant flag: =IF(Closing < 20L, "BREACH", "OK")
```

Collections come from the debtor book (60-day DSO means this quarter I'm largely collecting on prior-quarter sales), so I lag them: `=SUMIFS(Invoices, DueWeek, Wn)`. I use ~Rs 23 lakh/week baseline collections, dipping in the weeks reflecting the Q1 volume miss.

Step 3 — load the lumpy outflows. The killers in a direct forecast are the non-smooth items: supplier catch-up Rs 15 L (W4), quarterly-ish GST clusters, monthly payroll (last working week of each month), TDS (by the 7th → W1, W5, W10), loan EMI, and advance tax.

Loan EMI on Rs 1.20 cr @ 9.5% — interest alone is `1.20cr × 9.5% / 12 ≈ Rs 95,000/month`; with principal the EMI is ~Rs 2.6 lakh/month (assume ~5-yr amortisation). I place it in the first week of each month.

The deliverable

13-Week Direct Cash-Flow Forecast — NTSPL (Rs lakh), w/c 6 Jul → 28 Sep 2026

Week	Open	Collect	Supplier pay	Payroll	GST/TDS	EMI	Capex/Tax	Close	Covenant
W1	35.0	23	−16	0	−1.5	−2.6	0	37.9	OK
W2	37.9	22	−17	0	0	0	0	42.9	OK
W3	42.9	21	−16	0	−4 (GST)	0	0	43.9	OK
W4	43.9	22	−16	−9	0	0	−15 (supplier catch-up)	25.9	OK
W5	25.9	23	−17	0	−1.5	−2.6	0	27.8	OK
W6	27.8	20	−15	0	0	0	0	32.8	OK
W7	32.8	21	−16	0	−4 (GST)	0	0	33.8	OK
W8	33.8	22	−16	−9	0	0	0	30.8	OK
W9	30.8	23	−17	0	−1.5	−2.6	0	32.7	OK
W10	32.7	22	−16	0	−1.5 (TDS)	0	0	37.2	OK
W11	37.2	21	−16	0	−4 (GST)	0	0	38.2	OK
W12	38.2	22	−16	−9	0	0	−20 (adv tax)	15.2	**BREACH**
W13	15.2	24	−16	0	0	−2.6	0	20.6	OK (thin)

Analyst commentary: *Cash holds above the Rs 20 lakh covenant through most of the quarter, with the first squeeze in W4 (supplier catch-up drops us to Rs 25.9 L) and a covenant breach in W12 at Rs 15.2 L — the* *collision of month-end payroll and the 15-Sep advance-tax instalment. W13 recovers to Rs 20.6 L only because collections tick up. This is a timing problem, not a solvency problem — NWC is healthy at Rs 1.85 cr — but the bank tests the balance, so we must act before W12.*

Levers to free cash (ranked, with the W12 fix):

Lever	Mechanism	Cash impact	Fixes W12?
Accelerate collections (DSO 60→50)	Dunning top 10 debtors, early-pay 1% discount	~Rs 33 L freed	Yes — pull ~Rs 10–15 L into W11–12
Stagger advance tax / phase supplier	Move Rs 15 L supplier catch-up W4→W6; pay adv tax in two tranches	Smooths the two dips	Yes
Extend DPO 45→55 with key vendors	Negotiate net-55 terms	~Rs 33 L freed	Partial
Trim inventory (DIO 40→32)	Order in smaller lots on slow SKUs	~Rs 26 L freed	Partial
Draw the CC/OD limit	Bank overdraft as a buffer week	Bridges W12	Yes (last resort)

Recommendation: pull collections forward on the top 10 debtors and split the advance-tax payment — that alone lifts W12 back above Rs 20 L without touching the OD line.

How it's reviewed

The CFO/controller checks: (1) **roll-forward integrity** — each week's opening = prior week's closing, no gaps (a single broken link corrupts every downstream week). (2) **Covenant flag actually fires** — the `IF < 20L` must catch W12; a forecast that hides a breach is worse than useless. (3) **Lumpy items present** — payroll, EMI, GST, TDS, advance tax all placed on the *right* week, not smoothed away. (4) **CCC ties to balances** — 55 days ≈ Rs 1.85 cr NWC. (5) **Direct, not indirect** — this is actual receipts/payments (treasury view), not a P&L-derived accrual cash flow. (6) **Collections lag DSO** — you're collecting old sales, so the Q1 miss hits *these* weeks with a ~60-day delay.

Common mistakes & red flags

- **Recognising cash when revenue is booked** — with 60-day DSO, a sale today is cash ~8 weeks out. Putting revenue into the collection week overstates near-term cash badly.
- **Smoothing lumpy outflows** — averaging payroll/GST/tax across weeks hides the exact week you breach. The whole point of 13-week is the *timing*.
- **Sign errors / broken carry-forward** — one wrong link and the covenant check lies.
- **Confusing the CCC sign on DPO** — DPO is *subtracted*; a longer DPO shortens the cycle (good for cash).
- **Treating an overdraft draw as "found cash"** — it's a financing plug and carries interest; it's a last-resort lever, not a fix.
- **Ignoring GST input credit** — forecasting gross output GST without netting input credit overstates the outflow.

On the job & in the interview

The 13-week (a "flash" or "rolling") cash forecast is the treasury heartbeat of an FP&A team — it answers "will we make payroll and stay onside the covenant?" while the annual budget answers "will we hit PAT?" The **cash conversion cycle** is the structural story behind it: DSO + DIO − DPO. Improving CCC is often the cheapest financing available — freeing Rs 33 lakh by collecting 10 days faster beats drawing a loan at 9.5%.

Q: "Walk me through the cash conversion cycle and what a 55-day cycle costs us." *A: CCC = DSO 60 + DIO 40 − DPO 45 = 55 days — the gap between paying suppliers and collecting from customers. At Rs 12 cr revenue, ~Rs 3.3 lakh is tied up per day of cycle, so 55 days locks ~Rs 1.8 cr of operating cash, matching our Rs 1.85 cr net working capital. Every 10 days we shave frees ~Rs 33 lakh — real, interest-free financing.*

Q: "Your W12 breaches the Rs 20 lakh covenant. What do you do — and don't say 'draw the overdraft'?" *A: It's a timing collision of month-end payroll and the 15-Sep advance-tax instalment, not insolvency. First lever is collections: dunning the top 10 debtors and a small early-pay discount pulls Rs 10–15 lakh into W11–12. Second, I'd split the advance-tax payment and shift the W4 supplier catch-up to W6 to de-cluster the outflows. That restores the buffer without financing cost. The OD line is the last resort because it carries 9.5%+ interest and signals stress to the bank.*

Q: "Why a direct 13-week forecast and not just the indirect cash flow from the model?" *A: The indirect statement starts from PAT and adjusts for non-cash and working-capital movements — great for the annual view, but it's accrual-based and monthly. Treasury needs actual receipts and payments by week to manage the covenant and payroll. The direct method lists real cash in and out, so it catches the exact week we dip — which a monthly accrual view would completely miss.*

Building the Power BI Dashboard Behind the Pack

The ask

It's **14 July 2026**. Your MIS pack for Q1 (the Excel workbook with the P&L, the variance bridge, and a cover memo) went out last week and the CFO liked it. But this morning she forwards you an email from the Managing Director:

> "The Excel pack is fine for the finance team, but I want to *see* the numbers on my phone. Revenue vs budget, margin trend, top variances — one screen, refreshes itself. The board meets on the 28th. Can you build it in Power BI?"

The CFO adds one line: "*Same numbers as the pack. If the dashboard and the workbook ever disagree, we look like amateurs.*"

So the deadline is **Fri 25 July** (dry-run with CFO), board on the 28th. The real job: take the same actuals + budget that feed the Excel MIS and turn them into a Power BI report with a proper data model — not a pile of visuals stuck on a flat table.

What you're given

Two data sources, both exports you already use for the pack.

1. Actuals — from Tally, exported monthly as a "GL transaction" CSV (`Fact_Actuals`). One row per posting, already tagged with an account code and a segment:

Date	AccountCode	Segment	Amount (Rs)
2026-04-30	4000 (Revenue)	Goods	3,00,00,000
2026-04-30	4000 (Revenue)	Services	1,00,00,000
2026-04-30	5000 (COGS)	Goods	-2,25,00,000
...	...	...	...

2. Budget — the FY2026-27 annual budget, phased to months (`Fact_Budget`), same grain:

Month	AccountCode	Segment	Amount (Rs)
2026-04	4000	Goods	3,00,00,000
2026-04	4000	Services	1,00,00,000
...	...	...	...

Reconciliation anchors both must hit for the full year: **Revenue Rs 12.00 cr** (Goods 9.00 + Services 3.00), **COGS Rs 8.40 cr**, **Gross Profit Rs 3.60 cr (30%)**, EBIT Rs 1.596 cr, PAT ~Rs 1.11 cr. Q1: budget revenue **Rs 2.85 cr**, actual **Rs 2.70 cr**.

Build it — step by step

Step 1 — model shape: star schema, not one fat table

The rookie move is to load one wide table and slice it. The professional move is a **star schema**: two skinny *fact* tables (Actuals, Budget) surrounded by shared *dimension* tables. Dims are what you filter and group by; facts hold the numbers.

```
              ┌───────────┐
              │  Dim_Date │
              │ Date, Month,│
              │ Qtr, FY   │
              └─────┬─────┘
                    │ (1)
          ┌─────────┼──────────┐
          │ (*)            (*) │
   ┌──────┴──────┐      ┌──────┴───────┐
   │ Fact_Actuals│      │  Fact_Budget │
   │ Date,Acct,  │      │ Date,Acct,   │
   │ Segment,Amt │      │ Segment,Amt  │
   └──────┬──────┘      └──────┬───────┘
          │ (*)             (*)│
     ┌────┴─────┐  ┌─────────┐ │
     │Dim_Account│ │Dim_Segment├─┘
     │Code,Name, │ │ Goods /  │
     │Group,Sort │ │ Services │
     └──────────┘  └─────────┘
```

- **Dim_Date** — a proper calendar from 01-Apr-2026 to 31-Mar-2027, with columns `Date, MonthName, MonthNo, Quarter (Q1..Q4), FY`. Mark it as the date table.
- **Dim_Account** — maps code 4000 → "Revenue" → group "Income", 5000 → "COGS" → group "COGS", etc., plus a `SortOrder` so the P&L lines show in P&L order, not alphabetically.
- **Dim_Segment** — Goods / Services.

Both facts join to all three dims on a **single-direction, one-to-many** relationship (the "1" side is the dim). Two fact tables sharing dims is the classic budget-vs-actual pattern; the shared dims are what let one slicer filter both.

Step 2 — load & shape in Power Query

Get Data → CSV for each. In Power Query: - Set data types (Date as Date, Amount as Decimal). - In `Fact_Budget`, the month is text `2026-04`; convert to a real date (first of month) so it joins to Dim_Date. - Build `Dim_Date` with `= List.Dates(#date(2026,4,1), 365, #duration(1,0,0,0))` then add the Quarter/FY columns. - Build `Dim_Account` as a small entered table (7-8 rows) — faster than deriving it.

Step 3 — the DAX measures

Everything downstream is measures, never calculated columns on the facts. Base measures first, then everything is built from them.

```
Actual = SUM ( Fact_Actuals[Amount] )

Budget = SUM ( Fact_Budget[Amount] )

Variance = [Actual] - [Budget]

Variance % =
DIVIDE ( [Variance], [Budget] )        -- DIVIDE avoids /0 errors
```

Because COGS is stored as a negative, a favourable/unfavourable sign flip matters — so for cost/expense lines you want a "signed for the P&L" view. Two clean ways: store costs negative and just SUM (P&L nets naturally), or add a variance-direction helper. Keep it simple: costs negative, revenue positive, `[Actual]` sums the true P&L line.

```
Revenue =
CALCULATE ( [Actual], Dim_Account[Group] = "Income" )

Gross Profit =
CALCULATE ( [Actual], Dim_Account[Group] IN { "Income", "COGS" } )

GM % =
DIVIDE ( [Gross Profit], [Revenue] )

Revenue Budget =
CALCULATE ( [Budget], Dim_Account[Group] = "Income" )

Revenue Var % =
DIVIDE ( [Revenue] - [Revenue Budget], [Revenue Budget] )
```

Time intelligence — YTD and prior-period, which the board always wants:

```
Actual YTD =
TOTALYTD ( [Actual], Dim_Date[Date], "31-03" )   -- fiscal year-end 31 Mar

Revenue YTD =
TOTALYTD ( [Revenue], Dim_Date[Date], "31-03" )

Budget YTD =
TOTALYTD ( [Budget], Dim_Date[Date], "31-03" )
```

A conditional-format helper so unfavourable variances go red automatically:

```
Variance Colour =
VAR v = [Variance]
RETURN
IF ( v < 0, "#C0392B", "#1E8449" )   -- red if below budget, green if above
```

That's 10 measures. Notice none of them hard-code "Q1" or a segment — the slicers and the row context of each visual supply that, which is exactly why the star schema pays off.

Step 4 — sanity tie to Excel

Before building visuals, drop `[Revenue]` and `[Revenue Budget]` into a card with the Quarter slicer on Q1. You must see **Rs 2.70 cr** actual and **Rs 2.85 cr** budget. `[GM %]` on Q1 must read **28.5%**. Full-year `[Revenue]` = **12.00 cr**, `[Gross Profit]` = **3.60 cr**. If any of these are off, the model is wrong — fix it now, not after 12 charts are built.

The deliverable

Two report pages.

Page 1 — "Board Summary" (the MD's phone screen):

Zone	Visual	Measures
Top KPI row	4 cards	Revenue, Var %, GM %, EBIT (all YTD)
Left	Clustered column: Actual vs Budget by month	Actual, Budget
Right	Line: GM % trend by month vs 30% target line	GM %
Bottom	Bar: Variance by P&L line (sorted, red/green)	Variance + Variance Colour
Filters	Slicers: Quarter, Segment	—

Page 2 — "P&L Detail": a matrix — rows = Dim_Account (sorted by SortOrder), columns = Actual, Budget, Variance, Variance %, with the Quarter/Segment slicers carried across via *Sync slicers*.

Commentary in analyst voice, pinned as a text box: "*Q1 revenue Rs 2.70 cr, Rs 15 lakh (-5%) below budget, driven by a goods-volume miss (21,000 vs 22,500 units) partly cushioned by firmer pricing (ASP Rs 1,020 vs 1,000). Gross margin 28.5% vs 30% on mix and input cost. Services on plan. FY guidance unchanged pending Q2.*"

Set a **scheduled refresh** (Power BI Service, 8am daily) pointed at the same CSV/gateway folder finance drops the Tally export into — so it self-updates, which is the whole point of the ask.

How it's reviewed

The CFO/controller checks: 1. **Tie-out first.** Card totals must equal the Excel pack to the rupee — Q1 Rs 2.70 cr, GM 28.5%, FY 12.00 cr. This is the trust test. 2. **Sign logic.** Is an over-budget *cost* shown red (bad) not green? Revenue below budget red? Get the direction right per line. 3. **Slicer behaviour.** Pick "Services" — does *every* visual respond, and does the number reconcile (Services Rs 3.00 cr FY, on plan in Q1)? 4.

Blank/zero handling. Months with no data show blank, not error; DIVIDE prevents #DIV/0. 5. **Refresh actually works** end-to-end from the gateway, not just on your laptop.

Common mistakes & red flags

- **One flat table.** Works for 3 charts, collapses when you add budget, prior year, or a second segment. Star schema from day one.
- **Calculated columns instead of measures.** A column can't respond to a slicer's filter context the way a measure does, and it bloats the model.
- **Bi-directional relationships "to make it work."** Almost always a sign the model is wrong; keep single-direction dim→fact.
- **Two date columns, no Dim_Date.** Time intelligence (TOTALYTD) needs one marked date table shared by both facts.
- **Not using DIVIDE** — a single zero-budget line throws #DIV/0 and the whole visual errors on the board screen.
- **Dashboard disagreeing with the pack** by even a rupee — instant credibility loss. Tie out before you style anything.

On the job & in the interview

The "why": a dashboard isn't decoration — it's the *same governed numbers* in a self-refreshing, drill-able form. FP&A owns the semantic layer (the measures), so the definition of "Gross Margin %" lives in one place and can't drift. Jargon to own: **star schema, fact vs dimension, grain, filter context, measure vs calculated column, semantic model, RLS (row-level security), scheduled refresh**.

Q: "Why a star schema and not just one table?" A: "Facts hold the numbers at a single grain; dims hold the things I filter and group by. Separating them lets one slicer — say Segment — filter both my actuals and budget facts through shared dimensions, keeps the model small, and makes DAX behave predictably via filter context. A flat table can't cleanly do actual-vs-budget or time intelligence."

Q: "Walk me through a variance measure in DAX." A: "Base measures `Actual = SUM(Fact_Actuals[Amount])` and `Budget = SUM(Fact_Budget[Amount])`, then `Variance = [Actual]-[Budget]` and `Variance % = DIVIDE([Variance],[Budget])`. DIVIDE handles zero budgets gracefully. Every visual reuses these, so Q1's -Rs 15 lakh / -5% falls straight out of the Quarter slicer's filter context — I never hard-code the period."

Q: "How do you make sure it matches the Excel MIS?" A: "I tie out before styling: cards for Revenue, GM%, EBIT at Q1 and full year against the pack — Rs 2.70 cr, 28.5%, and the FY 12.00 cr / 3.60 cr anchors. If a number's off, the model's wrong. A dashboard that disagrees with the pack is worse than no dashboard."

The "By EOD" Ad-Hoc Data Request

The ask

It's **3:55pm on Tuesday, 21 July 2026**. You're heads-down on the Q2 forecast when the CFO stops at your desk on her way to a meeting:

> "Sales director thinks a few big accounts are eating our margin *and* paying late. Before I sit with him at 6, get me the **top 20 customers by gross margin**, and flag which ones are slow-paying. One page. By EOD."

No spec, no template, two hours. This is the bread-and-butter FP&A ad-hoc: pull it fast, get it *right*, present it in three lines the CFO can read walking into a room. You'll hit the ERP (SAP/Tally sits on a SQL database) directly, with Excel and Python as backups depending on where the data actually lives.

What you're given

Three tables in the reporting database (read replica of the ERP):

`sales` — one row per invoice line:

invoice_id	customer_id	invoice_date	revenue	cogs
INV-1001	C-014	2026-04-12	4,20,000	3,15,000
INV-1002	C-003	2026-04-15	1,80,000	99,000
...	...	...	...	...

`customer` :

customer_id	customer_name	segment
C-014	Deccan Switchgear	Goods
C-003	Krishna AMC Corp	Services

`payments` — settlements against invoices:

invoice_id	paid_date	amount
INV-1001	2026-06-30	4,20,000

Company context: **DSO target 60 days**. Anything materially over 60 is "slow-paying." Total FY revenue reconciles to **Rs 12.00 cr**; blended gross margin **30%** — so any single customer far below 30% is a margin drag worth the CFO's attention.

Build it — step by step

The SQL query (primary route)

One statement: join the three tables, aggregate per customer, compute margin and average days-to-pay, rank, take the top 20 by *margin rupees*.

```
WITH cust AS (
    SELECT
        s.customer_id,
        c.customer_name,
        SUM(s.revenue)                        AS revenue,
        SUM(s.revenue - s.cogs)               AS gross_margin_rs,
        SUM(s.revenue - s.cogs) * 1.0
            / NULLIF(SUM(s.revenue), 0)        AS gm_pct,
        AVG( DATEDIFF(day, s.invoice_date, p.paid_date) ) AS avg_days_to_pay
    FROM sales s
    JOIN customer c  ON c.customer_id = s.customer_id
    LEFT JOIN payments p ON p.invoice_id = s.invoice_id   -- LEFT: unpaid still count
    GROUP BY s.customer_id, c.customer_name
)
SELECT
    RANK() OVER (ORDER BY gross_margin_rs DESC) AS rnk,
    customer_name,
    revenue,
    gross_margin_rs,
    ROUND(gm_pct * 100, 1)          AS gm_pct,
    ROUND(avg_days_to_pay, 0)       AS days_to_pay,
    CASE WHEN avg_days_to_pay > 60 THEN 'SLOW' ELSE 'OK' END AS pay_flag
FROM cust
ORDER BY gross_margin_rs DESC
LIMIT 20;                          -- SQL Server: use SELECT TOP 20 instead
```

The load-bearing bits: - **`LEFT JOIN payments`** — an inner join would silently drop unpaid invoices, understating exposure. The slow-payers are exactly the ones you must not lose. - **`NULLIF(SUM(revenue),0)`** — guards the margin division against a zero-revenue customer. - **`RANK() OVER (ORDER BY gross_margin_rs DESC)`** — the window function does the ranking in-query; no eyeballing. - Rank by **margin rupees**, not margin %, because the CFO cares about who moves the P&L — a 15% GM customer doing Rs 80 lakh matters more than a 60% GM customer doing Rs 2 lakh. - Unpaid invoices push `avg_days_to_pay` up correctly only if you also treat NULL `paid_date` as "still open" — for a sharper version, `COALESCE(p.paid_date, CURRENT_DATE)` ages open invoices to today.

The Excel PivotTable alternative

If the ERP only gives you a dumped invoice register (`.xlsx`), no SQL access: 1. Add helper columns on the raw sheet: `Margin = revenue - cogs`; `DaysToPay = paid_date - invoice_date` (blank →

`=IF(paid_date="", TODAY()-invoice_date, paid_date-invoice_date)`). 2. PivotTable: **Rows** = customer_name; **Values** = Sum of Revenue, Sum of Margin, Average of DaysToPay. 3. Add a calculated **GM%** column next to the pivot: `=Margin / Revenue`. 4. Sort the pivot by Sum of Margin descending, **Top 20** via Value Filters → Top 10/20. 5. Flag column: `=IF(DaysToPay>60,"SLOW","OK")`, conditional-format SLOW red.

Equivalent single-cell pull without a pivot: `=SUMIFS(Margin, CustID, C-014)` and `=AVERAGEIFS(DaysToPay, CustID, C-014)`.

The Python / pandas version (repeatable / big file)

When it's a recurring ask or the file is too big for Excel to be pleasant:

```
import pandas as pd

sales    = pd.read_csv("sales.csv", parse_dates=["invoice_date"])
cust     = pd.read_csv("customer.csv")
pay      = pd.read_csv("payments.csv", parse_dates=["paid_date"])

df = (sales
      .merge(cust, on="customer_id", how="left")
      .merge(pay[["invoice_id", "paid_date"]], on="invoice_id", how="left"))

df["margin"] = df["revenue"] - df["cogs"]
df["days_to_pay"] = (df["paid_date"].fillna(pd.Timestamp("2026-07-21"))
                     - df["invoice_date"]).dt.days

g = (df.groupby("customer_name")
       .agg(revenue=("revenue", "sum"),
            margin=("margin", "sum"),
            days_to_pay=("days_to_pay", "mean"))
       .assign(gm_pct=lambda x: (x.margin / x.revenue * 100).round(1),
               pay_flag=lambda x: (x.days_to_pay > 60).map({True: "SLOW", False: "OK"}))
       .sort_values("margin", ascending=False)
       .head(20)
       .round({"days_to_pay": 0}))

g.to_excel("top20_margin.xlsx")
print(g)
```

Same logic as the SQL, in nine lines, and it re-runs next month by dropping in fresh CSVs.

The deliverable

Top of a single page, the table (extract — top rows shown):

Rank	Customer	Revenue (Rs)	Margin (Rs)	GM %	Days to pay	Flag
1	Deccan Switchgear	82,00,000	14,80,000	18.0%	78	SLOW
2	Godavari Controls	61,50,000	24,60,000	40.0%	44	OK
3	Krishna AMC Corp	55,00,000	24,75,000	45.0%	52	OK
4	Vijaya Electricals	71,00,000	12,78,000	18.0%	71	SLOW
…	…	…	…	…	…	…

Then the **three lines** the CFO actually reads:

> - **Top 20 customers = ~Rs 9.1 cr revenue (76% of book), ~Rs 2.6 cr margin.** Concentrated book.
> - **Margin drag:** Deccan and Vijaya together do ~Rs 1.5 cr revenue at **18% GM** vs our 30% blend — biggest pull on blended margin.
> - **These same two are the slowest payers (78 & 71 days vs 60 target)** — high revenue, thin margin, *and* slow cash. That's the account to renegotiate first.

That last sentence is the whole value: the CFO walks into the 6pm meeting knowing not just the data but the *action*.

How it's reviewed

- **Does the total tie?** Sum of all-customer margin should reconcile toward the 30% blended margin on Rs 12 cr; if your top-20 margin is implausibly high/low, a join fanned out or dropped rows.
- **Join integrity** — did a customer appear twice (duplicate in `customer`)? Did unpaid invoices vanish (inner join instead of left)?
- **Rank basis** — did you rank by margin Rs, as asked ("by margin"), not by revenue or GM%?
- **Slow-pay definition stated** — 60-day DSO target named on the page, so "SLOW" isn't arbitrary.
- **One page, plain English.** The CFO won't read the query; she reads the three lines.

Common mistakes & red flags

- **Inner join on payments** dropping unpaid (the slowest) invoices — understates exposure, hides the very accounts asked about.
- **Ranking by GM%** — surfaces tiny high-margin accounts and buries the big margin-rupee drivers.
- **Averaging averages** — computing DSO per invoice then averaging the percentages instead of the days; average days-to-pay is fine, but never average an already-averaged ratio.
- **Divide-by-zero** on a credit-note-only customer (negative/zero revenue) — guard with NULLIF/ `fillna`.
- **No reconciliation** — handing over a number that doesn't roll up to the Rs 12 cr / 30% anchors; always sanity-check the total.
- **Over-building** — a 40-line script when a PivotTable answers it in 10 minutes. Match the tool to a 2-hour deadline.

On the job & in the interview

The "why": ad-hoc requests are ~30% of an FP&A analyst's week. The skill isn't just SQL — it's turning a vague question into the *right* cut, reconciling it, and compressing it to a decision. Jargon: **JOIN types, GROUP BY, window function (RANK/ROW_NUMBER), fan-out, DSO, margin concentration, the three-line summary.**

Q: "You have a sales table and a payments table. How do you find slow-paying customers, and why LEFT JOIN?" A: "Left join sales to payments on invoice_id so unpaid invoices survive — they're precisely the slow/non-payers. Compute days-to-pay as paid_date minus invoice_date, coalescing NULL to today to age open items, group by customer, average it, and flag anything over our 60-day DSO target. An inner join would silently drop the worst offenders."

Q: "Rank the top 20 by margin — SQL?" A: "Aggregate `SUM(revenue-cogs)` per customer in a CTE, then `RANK() OVER (ORDER BY margin DESC)` and take 20. I rank by margin rupees, not %, because the CFO cares about P&L impact — a big low-margin account outranks a tiny high-margin one."

Q: "It's 4pm, due at 6. Excel or Python?" A: "For a one-off, whichever gets a correct, reconciled answer fastest — usually a PivotTable if I already have the dump. I'd only reach for Python/SQL if it's recurring or the file's too big for Excel. Speed and a clean tie-out beat elegance on a two-hour deadline."

The FP&A Interview Simulation + Cheat-Sheet

The ask

It's **Monday, 27 July 2026**. A friend forwards a role: *FP&A Analyst, mid-market manufacturer, Hyderabad.* The JD reads almost exactly like your Nirvana job — annual budget, monthly variance, forecasting, a bit of capex appraisal, Excel + Power BI. You have a first-round with the FP&A Manager on Thursday and the CFO round after.

This chapter *is* your prep: a realistic mock, using the same Nirvana numbers you've been living in, followed by a one-page cheat-sheet to skim in the cab.

What you're given

The Nirvana anchors, which you'll quote from memory in the room: **Revenue Rs 12.00 cr** (Goods 9.00 = 90,000 units x Rs 1,000; Services 3.00 = 250 AMCs x Rs 1,20,000). **COGS 8.40 cr, GP 3.60 cr (30%)**. Opex: Employee 1.08 + Other 0.78 + Depreciation 0.144. **EBIT 1.596 cr; Finance cost 0.09; PBT 1.506; PAT ~1.11 cr** (25%+cess). Q1: budget 2.85 cr, actual 2.70 cr (-15 lakh / -5%), GM 28.5% vs 30%. Capex case: Rs 40 lakh, saves Rs 12 lakh/yr x 6, salvage Rs 4 lakh, 12% discount. Working capital: DSO 60, DPO 45, DIO 40.

Build it — the mock interview

Q1 — "Walk me through the variance between your Q1 budget and actual." "Q1 revenue was Rs 2.70 cr against a Rs 2.85 cr budget — Rs 15 lakh unfavourable, -5%. It's a goods story: volume came in at 21,000 units versus 22,500 budgeted, a ~Rs 15 lakh volume miss at standard price, partly offset by price — ASP realised Rs 1,020 vs Rs 1,000 budget, ~+Rs 4 lakh. Services were on plan. Below the line, gross margin was 28.5% vs 30% on adverse mix and input cost, and opex ran Rs 5 lakh over on two early hires. So: volume-led revenue miss, cushioned by price, with a margin squeeze on top."

Q2 — "Split that into a price/volume bridge. Formulas?" "Volume variance = (actual units - budget units) x budget price = (21,000 - 22,500) x Rs 1,000 = -Rs 15 lakh. Price variance = (actual price - budget price) x actual units = (1,020 - 1,000) x 21,000 = +Rs 4.2 lakh. Net goods variance ≈ -Rs 10.8 lakh; the rest of the -15 is mix/services timing. I always cost the volume variance at *budget* price and the price variance at *actual* volume, so they don't double-count."

Q3 — "Budget vs forecast — what's the difference?" "The budget is the fixed annual target set once, pre-year — my scorecard. The forecast is the latest expected outturn, re-cut each quarter using actuals to date. After a -5% Q1, budget still says Rs 12 cr, but my forecast might say Rs 11.7 cr if the volume softness persists. Budget stays frozen so variances stay honest; the forecast is what management steers by."

Q4 — "Walk me through the three statements and how they link." "P&L ends in net income. That net income flows to the balance sheet via retained earnings, and it's the starting line of the cash flow statement. Cash flow adjusts net income for non-cash items — add back the Rs 14.4 lakh depreciation — and for working-capital moves — a rise in debtors is a cash outflow. The closing cash from the cash flow becomes the cash line on the balance sheet. Depreciation reduces PP&E on the balance sheet; capex increases it. Everything ties: the balance sheet only balances if all three are wired correctly."

Q5 — "If depreciation goes up Rs 10 lakh, what happens across the three statements?" "P&L: EBIT and PBT drop Rs 10 lakh; at ~25% tax, PAT falls Rs 7.5 lakh. Cash flow: start from the lower PAT but add back the full Rs 10 lakh depreciation, so operating cash *rises* Rs 2.5 lakh — the tax shield. Balance sheet: PP&E down Rs 10 lakh, retained earnings down Rs 7.5 lakh, cash up Rs 2.5 lakh — and it still balances."

Q6 — "Walk me through a DCF." "Project free cash flows for a forecast horizon, discount each to today at WACC, add a terminal value for the years beyond, discount that back too, and sum for enterprise value. FCF = EBIT x (1-tax) + depreciation - capex - increase in working capital. Terminal value is usually Gordon growth — final-year FCF x (1+g) / (WACC-g) — or an exit multiple. Subtract net debt to get equity value. For Nirvana at ~12% WACC, the sensitivity is all in WACC and the terminal growth assumption."

Q7 — "Appraise our warehouse-automation capex: Rs 40 lakh, saves Rs 12 lakh/yr for 6 years, Rs 4 lakh salvage, 12%." "Cash flows: -40 now, +12 in years 1-5, +16 in year 6 (12 saving + 4 salvage). At 12%, the annuity factor for 6 years is 4.111, so 12 x 4.111 = Rs 49.3 lakh, plus salvage 4 x 0.507 = Rs 2.0 lakh, gross Rs 51.3 lakh, minus 40 = **NPV ≈ +Rs 11.3 lakh.** Positive, so accept. IRR is roughly 20% — comfortably above the 12% hurdle. Simple payback is 40/12 ≈ 3.3 years."

Q8 — "NPV vs IRR — which do you trust and why?" "NPV, when they conflict. IRR is intuitive as a percentage and fine for independent projects, but it assumes reinvestment at the IRR itself, can give multiple answers with unconventional cash flows, and is blind to scale — a 25% IRR on Rs 2 lakh loses to a 15% IRR on Rs 2 cr. NPV measures rupees of value added at the true cost of capital, so for mutually exclusive choices I rank on NPV."

Q9 — "Our cash conversion cycle?" "CCC = DIO + DSO - DPO = 40 + 60 - 45 = **55 days.** We fund 55 days of operations. The quickest lever is DSO — from the last exercise, our two biggest accounts pay at 70-78 days against a 60 target; tightening them pulls cash in without touching margin."

Q10 — "Live Excel test: monthly revenue by segment from a transaction dump — formula?" "`=SUMIFS(Amount, Segment, $F2, MonthCol, G$1)` with the segment down the rows and months across, mixed references locked so it fills both ways. For pulling a price or a customer attribute I'd use `XLOOKUP(id, id_range, return_range, "not found")` — exact match, no column-count fragility like VLOOKUP. NPV of the capex line: `=NPV(12%, year1:year6) + initial_outlay` where the outlay is a negative in the cell before the range."

Q11 — "How would you build next year's budget?" "Driver-based, bottom-up. Revenue isn't a percentage bump — it's units x ASP per segment: 90,000 units x Rs 1,000 for goods, 250 AMCs x Rs 1,20,000 for services. COGS off segment gross-margin assumptions (25% goods, 45% services). Opex built from headcount (15 to 18 by Q4) and a cost schedule. Then phase to months, layer the balance sheet and cash flow, and check the outputs — 30% GM, ~Rs 1.6 cr EBIT — are sane before locking."

Q12 — "Guesstimate: AMC market for industrial electrical AMCs in Hyderabad?" "Top-down. Say ~5,000 mid/large industrial and commercial sites in greater Hyderabad with meaningful electrical infrastructure; assume 60% outsource AMC = 3,000 contracts; average contract ~Rs 1.2 lakh (our own ASP) = ~Rs 36 cr addressable. We do 250 contracts / Rs 3 cr, so ~8% share — plausible for a regional player, and it says there's room to grow before the market caps us."

Q13 — "You find a formula error in the pack ten minutes before the board. What do you do?" "Quantify it first — is it material to a decision? Flag it to the CFO immediately with the corrected number and the impact; never let a wrong number go into the room to save face. Then fix the source, re-tie the totals to the anchors, and note what control failed so it doesn't recur. Integrity over ego — one silent wrong number costs more than one awkward correction."

Q14 — "Why FP&A and not audit or accounting?" "Accounting tells you what happened; FP&A asks what it means and what to do next. I like living in the model — connecting a volume miss to a forecast cut to a cash impact to a decision the CFO can act on. It's forward-looking and business-facing, which is where I want to be."

The deliverable — one-page FP&A cheat-sheet

Core formulas

Metric	Formula
Gross Margin %	Gross Profit / Revenue
Volume variance	(Actual units - Budget units) x **Budget** price
Price variance	(Actual price - Budget price) x **Actual** units
Contribution	Revenue - variable costs
EBIT	Revenue - COGS - Opex
NPV	$\Sigma\ CF_t / (1+r)^t$ - initial outlay
IRR	rate where NPV = 0
Payback	Outlay / annual cash inflow
WACC	E/V·Re + D/V·Rd·(1-tax)
CCC	DIO + DSO - DPO
DSO	Debtors / Revenue x 365
FCF	EBIT(1-tax) + Deprec - Capex - ΔWC
Terminal value	FCF(1+g) / (WACC-g)

FP&A annual calendar

When	Activity
Jan-Mar	Annual budget build (driver-based), board approval
Monthly (by day 5-7)	Close, actuals load, MIS pack, variance bridge
Quarterly	Re-forecast, board deck, reforecast vs budget
Ad-hoc	Capex appraisals, pricing, scenario/what-if
Year-end	True-up, next-year planning kickoff

Must-know tools

- Excel: `SUMIFS`, `XLOOKUP` / `INDEX-MATCH`, `NPV`, `IRR`, `PMT`, mixed refs `$A1` / `A$1`, data tables for sensitivity, PivotTables.

- DAX: `Actual = SUM( … )`, `Variance = [Actual]-[Budget]`, `DIVIDE()`, `TOTALYTD()`, `CALCULATE()`.
- SQL: `JOIN`, `GROUP BY`, `RANK() OVER( … )`, `CASE WHEN`.

KPI definitions to state crisply

- **Budget** = fixed annual target; **Forecast** = latest expected outturn; **Actual** = what happened.
- **Favourable/Unfavourable** — revenue above budget or cost below budget = favourable.
- **Gross margin** vs **EBIT margin** vs **PAT margin** — know all three for Nirvana: 30% / 13.3% / ~9.3%.
- **Accretive/dilutive, run-rate, YoY, YTD, variance bridge, waterfall.**

How it's reviewed

The interviewer is checking three things: (1) can you *reconcile* — do your numbers tie, do you quote the 30% GM and Rs 1.6 cr EBIT without fumbling; (2) do you connect the dots — variance → forecast → cash → decision, not isolated facts; (3) judgment under pressure — the error-before-the-board answer matters as much as the DCF. Precise numbers spoken with a "so what" beat vague theory every time.

Common mistakes & red flags

- Costing volume variance at actual price (double-counts with price variance) — use budget price.
- Confusing budget and forecast, or calling a re-forecast a "revised budget."
- Quoting IRR as always superior — misses scale and reinvestment flaws.
- A DCF walk-through that forgets to subtract net debt or hand-waves terminal value.
- Guesstimates with no stated assumptions — the number matters less than the structure.
- Rounding away the anchors — if you can't get back to Rs 12 cr / Rs 3.6 cr / Rs 1.11 cr, you don't know your own case.

On the job & in the interview

FP&A interviews reward the person who *lives in one model* and can speak it fluently — which is the whole point of running the Nirvana case end to end. Bring one printed page, know your anchors cold, and always answer a number question with the number *and* the "so what." The three questions they almost always ask: a variance walk-through, a three-statement linkage, and a capex/DCF — all three are answered above, in rupees, from your own book.